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Mysore
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**VTU Question Papers
“MECHANICAL ENGINEERING”**

Odd Semester

2022-23

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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CBCS SCHEME

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18ME51

Fifth Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023 Management and Economics

Time: 3 hrs.

Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Interest Factor table is permitted.

Module-1

- 1 a. Define Management and discuss its nature and characteristics. (06 Marks)
b. Discuss Fayol's principles of Administrative Management. (14 Marks)

OR

- 2 a. What is Planning? Explain different steps in Planning. (10 Marks)
b. What do you understand by term Planning Premises? Explain different types of Planning premises. (10 Marks)

Module-2

- 3 a. Briefly explain principles of Organisation. (10 Marks)
b. What is Recruitment? Explain sources of Recruitment. (10 Marks)

OR

- 4 a. List various Motivation theories. Explain Maslow need Hierarchy theory in brief. (10 Marks)
b. Explain requirements of a good control system. (10 Marks)

Module-3

- 5 a. Explain Laws of Supply and Demand using suitable sketch. (08 Marks)
b. With a neat sketch, explain Cash flow diagram. (04 Marks)
c. Determine the effective interest rate for nominal annual rate of 8% compounded.
i) Daily (Assume 365 days/yr) ii) Monthly iii) Quarterly iv) Semi – annually. (08 Marks)

OR

- 6 a. What is Law of Diminishing return? Write its limitations. (03 Marks)
b. Discuss terms : i) Price elasticity of demand ii) Income elasticity of demand. (08 Marks)
c. A person is planning for his retired life. He has 10 more years of service. He would like to deposit 20% of his salary, which is Rs 4000 in first year and thereafter he wishes to deposit amount with annual increase of Rs 500 for next nine years with an interest rate of 15%. What will be the maturity amount? (09 Marks)

Module-4

- 7 a. Following table gives initial outlay and annual revenue of a production firm using three various alternatives. Find the best alternative based on present worth if the rate of interest is 20% compounded annually. (09 Marks)

	Initial Outlay	Annual Revenue	Life (Years)
Alternative 1	13,00,000	4,00,000	10
Alternative 2	21,00,000	6,50,000	10
Alternative 3	23,00,000	8,60,000	10

- b. Find the most economical alternatives from following on the basis of equivalent future worth at interest rate of 9.5% per year.

Alternative 1 : Initial purchase cost = Rs 15,00,000 , Annual operating cost = Rs 35,000 starting from end of second year till end of life , Annual revenue generated = Rs 340000 for first 4 yrs then Rs 320000 afterwards till end of useful life. Expected salvage value is Rs 430000 and useful life = 8 yrs.

Alternative II : Initial purchase cost = Rs 1800000 , Annual operating cost = Rs 2500 , Annual revenue generated = Rs 365000 , Salvage value = Rs 550000 , Useful life = 8 yrs.

(11 Marks)

OR

- 8 a. Explain IRR , ERR and MARR. Enlist the misconcepts of IRR. (08 Marks)
 b. A firm has identified three mutually exclusive investment proposals whose details are given below. The life of three investments is estimated to be five years with negligible salvage value. The minimum rate of return for the firm is 12%. Find the best alternative based on rate of return method of comparison. (12 Marks)

	Alternative		
	A ₁	A ₂	A ₃
Investment	1,50,000	2,10,000	2,55,000
Annual net income	45,570	58,260	69,000

Module-5

- 9 a. With a block diagram, explain how a selling price of a product is determined? (08 Marks)
 b. The expenditure incurred in manufacturing machine is as follows :
 1) Material consumed = Rs 55,00,000 2) Indirect factory wages = Rs 8,00,000
 3) Directors fees = Rs 3,00,000 4) Cost of advertisement = Rs 1,00,000
 5) Net profit = Rs 1,20,000 6) Depreciation on sales dept car = Rs 11,000
 7) Printing and stationery cost = Rs 2500 8) Depreciation of plant = Rs 45,000
 9) Direct wages = Rs 6,50,000 10) Factory rent = Rs 60,000
 11) Telephone and postage charges = Rs 15,000
 12) Gas and electricity = Rs 50,000 13) Office salaries = Rs 2,10,000
 14) Office rent = Rs 50,000 15) Show room rent = Rs 1,50,000
 16) Sales man commission = Rs 26,500 17) Sales dept car expensed = Rs 15,000
 Determine i) Direct cost ii) Factory cost iii) Total cost of production
 iv) Cost of sales v) Selling price. (12 Marks)

OR

- 10 a. What do you mean by Depreciation? Discuss various causes of depreciation. (10 Marks)
 b. A Company has purchased on equipment whose first cost is Rs 2,00,000 with an estimated life of eight years. Estimated salvage value is 40,000 at the end of its life. Determine the depreciation charges and book value at the end of second year by sum of year's digit method of depreciation. (10 Marks)

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18ME52

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Design of Machine Elements – I

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. Use of design data handbook is permitted.

Module-1

- 1 a. Discuss the factors influencing the selection of suitable material for machine element. (08 Marks)
- b. Determine the safe load that can be carried by a bar of rectangular cross section shown in Fig.Q1(b). Limiting the maximum stress to 130 MPa taking stress concentration into account and assume thickness of bar as 10 mm.

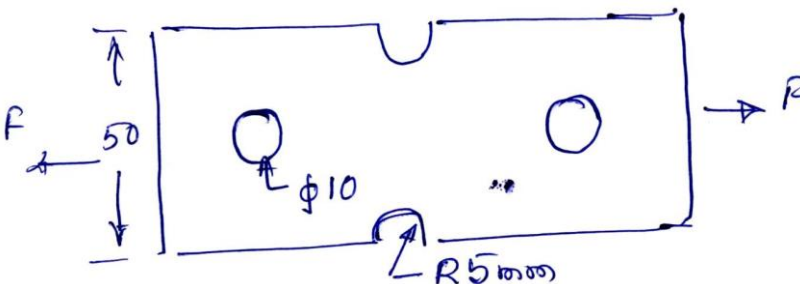


Fig.Q1(b)

(12 Marks)

OR

- 2 a. Explain the following theories of failure : (10 Marks)
 - (i) Maximum normal stress theory
 - (ii) Maximum shear stress theory
 - (iii) Distortion energy theory
- b. A machine element made of C45 steel is subjected to a system of loads, following stresses are induced at critical point:
 $\sigma_x = 150 \text{ MPa}$, $\sigma_y = 100 \text{ MPa}$ and $\tau_{xy} = 50 \text{ MPa}$
 Find the factor of safety according to:
 - (i) Maximum normal stress theory
 - (ii) Maximum shear stress theory
 - (iii) Distortion energy theory(10 Marks)

Module-2

- 3 a. Derive Soderberg's equation. (06 Marks)
- b. A hot rolled steel rod is subjected to torsional load that varies from +330 N-m clockwise to 110 N-m counter clockwise and an applied bending moment varies from +440 N-m to -220 N-m. The rod is of uniform cross section. Determine the required diameter rod. The material has an ultimate tensile strength of 550 MPa and yield strength of 410 MPa. Assume a factor of safety 1.5. Take the endurance limit as half of the ultimate strength. (14 Marks)

OR

- 4 a. List and explain the various factors effecting the endurance limit of the material. (08 Marks)
- b. An unknown weight falls through 20 mm as to a collar rigidly attached to the lower end of a vertical bar 2 meter long and 500 mm² section. If the maximum instantaneous extension is 2 mm. What is the corresponding stress and the value of unknown weight? Take $E = 200 \text{ GPa}$. (06 Marks)
- c. A cantilever beam of span 800 mm has a rectangular cross section of depth 200 mm. The free end of beam is subjected to a transverse load of 1 kN that drops on to it from a height of 40 mm. Selecting C40 steel as material and a factor of safety 2. Determine the width of rectangular cross section. Assume $E = 200 \text{ GPa}$. (06 Marks)

Module-3

- 5 A commercial shaft 1 metre long supported between bearings has a pulley of 600 mm diameter weighing 1 kN, driven by a horizontal belt drive keyed to the shaft at a distance of 400 mm to the left of the right bearing and receives 25 KW at 1000 rpm. Power from the shaft is transmitted from the 20° spur pinion of a pitch circle diameter 200 mm which is mounted at 200 mm to the right of the left bearing to a gear such that tangential force on gear acts vertically upwards. Take the ratio of the belt tension is 3. Determine the standard size of the shaft based on maximum shear stress theory. Assume $C_m = 1.75$, $C_t = 1.25$. (20 Marks)

OR

- 6 a. Compare weight, strength and stiffness of hollow shaft of same external diameter of that solid shaft. The inside diameter being half the external diameter. Both the shafts have same material and length. (06 Marks)
- b. Design a cast iron flanged coupling for a steel shaft transmitting 100 KW at 250 rpm. Take the allowable shear stress for the shaft as 40 N/mm². The angle of twist is not to exceed 1° in a length of 20 diameters. Allowable shear stress for the bolts is 13 MPa. The allowable shear stress in the flange is 14 MPa for the key is 40 MPa. Allowable compressive stress in key is 80 MPa. (14 Marks)

Module-4

- 7 a. Explain in detail various possible modes of failure of riveted joint. (06 Marks)
- b. Design a double riveted butt joint with two equal cover plates for the longitudinal seam of a boiler shell 1.5 m in diameter subjected to a steam pressure of 0.95 N/mm². Assume an efficiency of 75% allowable tensile stress in the plate of 90 N/mm², allowable crushing stress of 140 N/mm² and an allowable shear stress in the rivet of 50 N/mm². (14 Marks)

OR

- 8 a. A bracket having a load of 15 kN is to be welded as shown in Fig.Q8(a). Find the size of weld required, if allowable shear stress is not to exceed 80 N/mm².

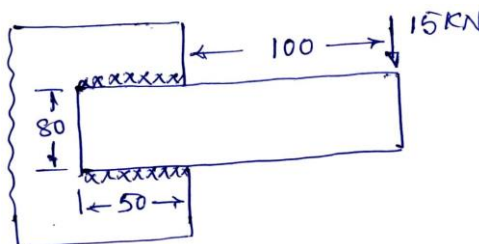


Fig.Q8(a)

(10 Marks)

- b. Determine the size of rivets required for the bracket shown in Fig.Q8(b). Take allowable shear stress of rivet material as 100 N/mm^2 .

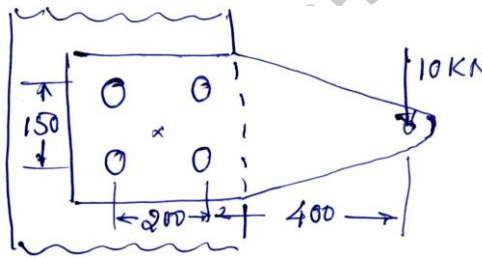


Fig.Q8(b)

(10 Marks)

Module-5

- 9 a. Obtain an expression for torque required to lift the load on a square threaded screw. (08 Marks)
- b. Design a socket and Spigot type cotter joint to sustain an axial load of 100 kN. The material selected for the joint has the following design stresses $\sigma_t = 100 \text{ N/mm}^2$, $\sigma_c = 150 \text{ N/mm}^2$ and $\tau = 60 \text{ N/mm}^2$. (12 Marks)

OR

- 10 a. Explain self locking and overhauling of power screw. (06 Marks)
- b. The cotter of a broaching machine is pulled by square threaded screw of 55 mm external diameter and 10 mm pitch. The operating nut takes the axial load of 400 N. On a flat surface of 60 mm and 90 mm internal and external diameters respectively. If the coefficient of friction is 0.15 for all contact surfaces, determine the power required to rotate the nut when the cutting speed is 6 m/min. Also find the efficiency of the screw. (14 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Dynamics of Machines

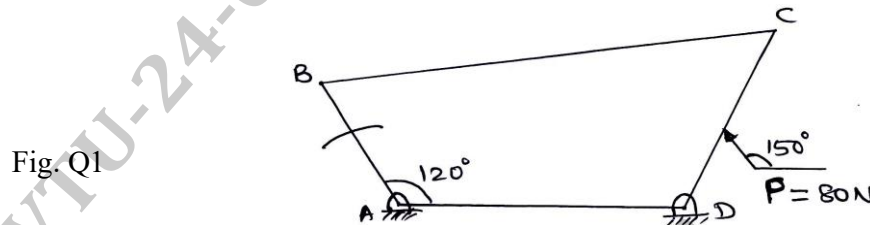
Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 A four bar mechanism with the following dimensions is acted upon by a force 80N, 150° on the link DC. Determine the input torque on the link AB for the static equilibrium of the mechanism for the given configuration Fig. Q1. AB = 400mm, BC = 1000mm, CD = 750mm and DE = 350mm, AD = 500mm. (20 Marks)



OR

- 2 a. State the condition of equilibrium of a body subjected to a system of
i) Two force ii) Three force iii) Two force and a torque. (06 Marks)
- b. In a vertical engine, the length of connecting rod is 4.5 times the crank. The mass of reciprocating parts is 120kg and the crank length is 220mm. The engine runs at 250 rpm. The load on the piston due to steam pressure is 25 kN, when the crank has turned through an angle of 120° from the top dead centre. Determine i) Net effective driving force on the piston ii) Thrust on connecting rod iii) Thrust on the bearings iv) Turning moment on the crank shaft. (14 Marks)

Module-2

- 3 a. Explain Static and Dynamic balancing of rotating masses. (04 Marks)
- b. Four masses A, B, C and D carried on a shaft at radii 100mm, 125mm, 200mm and 150mm respectively. The planes at which masses are rotating are placed 600mm apart. The mass B, C and D are 10kg, 5kg and 4kg respectively. Find the mass of A and relative angular position of the four masses so that the shaft will be in equilibrium. (16 Marks)

OR

- 4 The firing order in a six cylinder four stroke in line engine is 1 – 4 – 2 – 6 – 3 – 5. The piston stroke is 100mm and length of each connecting rod is 200mm. The pitch of the cylinder centre lines are 100mm, 100mm, 150mm, 100mm and 100mm respectively. The reciprocating mass per cylinder is 1kg and the engine runs at 3000 rpm. Determine the unbalanced primary and secondary forces and couples, if any. Take central plane of the engine as reference plane. (20 Marks)

Module-3

- 5 a. Derive the expression for speed of a Porter Governor with usual notations, taking friction into account. (08 Marks)

- b. The upper arms of a Porter Governor has lengths 350mm and are pivoted on the axis of rotation. The lower arms have lengths 300mm and are attached to the sleeve at a distance of 40mm from the axis. Each ball has a mass of 4 kg and mass on the sleeve is 45kg. Determine the equilibrium speed for a radius of rotation of 200mm and find the effort and power of governor for 1% speed change. (12 Marks)

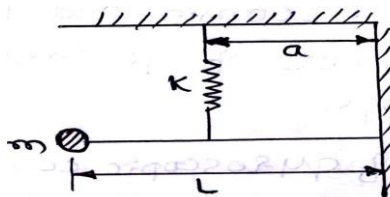
OR

- 6 a. Explain the effect of gyroscopic couple on an Aeroplane. (06 Marks)
 b. The turbine rotor of a ship has a mass of 3500kg. It has a radius of gyration of 0.45m and a speed of 3000 rpm clockwise when looking from stress. Determine the gyroscopic couple and its effect upon the ship.
 i) When the ship is steering to the left on a curve of 100m radius at a speed of 36km/hour.
 ii) When the ship is pitching with SHM the bow falling with its maximum velocity. The period of pitching is 40 sec and the total angular displacement between the two extreme position of pitching is 12° . (14 Marks)

Module-4

- 7 a. Define the following with respect to vibration : i) Degrees of freedom ii) Amplitude
 iii) Resonance iv) Natural frequency v) Damping factor. (10 Marks)
 b. Determine the natural frequency of the system shown in Fig. Q7(b) by Newton's and Energy method. (10 Marks)

Fig. Q7(b)



OR

- 8 a. Set up the differential equation for a spring mass damper system and obtain complete solution for the critically damped condition. (10 Marks)
 b. A vibrating system having a mass of 3kg, spring stiffness of 100 N/mm and damping coefficient of 3 N-S/m. Determine damping ratio, damped natural frequency, logarithmic decrement, ratio of two consecutive amplitudes and number of cycles after which the original amplitude is reduced to 20%. (10 Marks)

Module-5

- 9 a. Define "Transmissibility". Derive an expression for force transmissibility. (10 Marks)
 b. A 35kg block is connected to a support through a spring of stiffness 1.4×10^6 N/m in parallel with dashpot of damping coefficient 1.8×10^3 N-S/m. The support is given a harmonic displacement of amplitude 10mm at a frequency of 35Hz. Compute the steady state amplitude of the absolute displacement of the block. (10 Marks)

OR

- 10 a. Derive an equation for steady state amplitude for forced vibration with rotating unbalance. (10 Marks)
 b. A rotor has a mass of 12kg and is mounted midway on a 24mm diameter horizontal shaft supported simply at the ends by two bearings. The bearings are 1m apart. The shaft rotates at 2400 rpm. If the centre of mass of the rotor is 0.11mm away from the geometric centre of the rotor due to manufacturing defect, find i) the amplitude of the steady state vibration ii) the dynamic force transmitted to the bearing. Take $E = 200$ GPa. (10 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Turbomachines

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define specific speed of a turbine. Derive an expression for specific speed of a turbine. (08 Marks)
- b. A model of a centrifugal pump absorbs 5kW at a speed of 1500rpm. Pumping water against a head of 6m. The large prototype pump is required to pump water to a head of 30m. The scale ratio of diameter is 4. Assuming some efficiency and similarities, find the speed, power of prototype and ratio of discharge of prototype and model. (08 Marks)
- c. For power generating turbomachines, define
 - i) Total – to – total efficiency
 - ii) Total – to – static efficiency. (04 Marks)

OR

- 2 a. With usual notations, derive an expression for infinitesimal stage efficiency during compression process with an aid of T-S plot. (08 Marks)
- b. An air compressor has eight stages of equal pressure ratio 1.3. The flow rate through the compressor and its overall efficiency are 45Kg/s and 80% respectively. If the conditions of air at entry are 1 bar and 35°C, determine,
 - i) State of air at compressor exit
 - ii) Polytropic efficiency (06 Marks)
- c. Compare the turbomachine with positive displacement machines. (06 Marks)

Module-2

- 3 a. Derive alternate form of Euler's turbine equation and explain the significance of each energy component. (10 Marks)
- b. At a stage of an axial flow impulse turbine, the mean blade diameter is 80cm and the speed is 3000 rpm. The absolute velocity of the fluid at inlet is 300m/sec and is inclined at 20° to the plane of the wheel. If the utilization factor is 0.85 and the relative velocity at rotor exit is equal to that at inlet, determine
 - i) Inlet and exit blade angles
 - ii) Power output for a mass flow rate of 1Kg/sec. (10 Marks)

OR

- 4 a. Define degree of reaction for an axial flow machine. Prove that degree of reaction for an axial flow device assuming constant velocity of flow is given by

$$R = \frac{V_a}{2u} [\cot\beta_1 + \cot\beta_2]$$
(10 Marks)
- b. In a turbine stage with 50% reaction the tangential blade speed is 98.5 m/sec. The steam velocity at the nozzle exit is 155 m/sec and the nozzle angle is 18°. Assuming symmetric inlet and outlet velocity triangles. Compute the inlet blade angle for the rotor and power developed by the stage assuming a steam flow rate of 10Kg/sec. Also find the utilization factor. (10 Marks)

Module-3

- 5 a. Draw the inlet and exit velocity triangle for a single stage impulse steam turbine and prove that maximum blade efficiency is given by

$$\eta_{\text{bmax}} = \cos^2 \alpha_1$$

 Assume $v_{r1} = v_{r2}$ and $\beta_1 = \beta_2$ (10 Marks)
- b. The following particulars refer to a stage of a Parson's steam turbine. The mean diameter of the blade ring is 70cm, the steam velocity at the inlet of moving blades is 160m/sec, the outer blade angle of moving blade β_2 is 20° . The steam flow through the blades is 7Kg/sec, Speed 1500rpm and η_{st} is 0.8. Draw the velocity diagrams and find the following :
 i) Blade inlet angle
 ii) Power developed in the stage
 iii) Available isentropic enthalpy drop. (10 Marks)

OR

- 6 a. Define and explain nozzle efficiency and stage efficiency. (04 Marks)
 b. With a neat sketch, explain the velocity compounding. (06 Marks)
 c. In a stage of an impulse turbine provided with single row wheel, the mean diameters of the blade ring is 80cm and the speed of rotation is 3000rpm. The steam issues from the nozzle with a velocity of 300m/sec and the nozzle angle is 20° . The rotor blades are equiangular and blade velocity coefficient is 0.85. What is the power developed in the blades when the axial thrust on the blade is 140N? (10 Marks)

Module-4

- 7 a. Derive an expression for force, power and efficiency of a Pelton turbine assuming no frictional losses with the help of velocity triangles. (10 Marks)
 b. The following data is given for a Francis turbine net head = 70m, Speed = 600rpm, Shaft power = 368kW, $\eta_0 = 86\%$, $\eta_h = 95\%$, flow ratio = 0.25, breadth ratio = 0.12, outer diameter of runner = 2 times inner diameter of runner, velocity of flow is constant at inlet and outlet, the thickness of vanes occupies 10% of the circumferential area of the runner and discharge is radial at outlet. Determine :
 i) Guide blade angle
 ii) Runner vane angles at inlet and outlet
 iii) Diameters of runner at inlet and outlet
 iv) Width of the wheel at inlet (10 Marks)

OR

- 8 a. Draw the cross sectional views of a Kaplan turbine and explain its working with a neat sketches of velocity triangles at inlet and outlet of Kaplan turbine runner. (10 Marks)
 b. A three – jet Pelton wheel is required to generate 10,000kW under a head of 400m. The blade angle at outlet is 15° and reduction in relative velocity over the bucket is 5%. If the overall efficiency is 80%, $C_v = 0.98$ and speed ratio = 0.46. Find
 i) Diameter of jet
 ii) Total flow in m^3/sec
 iii) Force exerted by a jet on the buckets (10 Marks)

Module-5

- 9 a. Applying Bernoulli's equation between the inlet and exit of the impeller of a centrifugal pump. Show that the static pressure rise is given by,

$$(P_2 - P_1) = \rho/2 \left[v_{f1}^2 + u_2^2 - v_{f2}^2 \cos^2 \beta_2 \right]$$

 Where, v_{f1} = Velocity of flow at inlet
 v_{f2} = Velocity of flow at exit
 β_2 = Blade angle at exit
 u_2 = Blade speed at exit
 ρ = density of fluid (08 Marks)
- P_1 and P_2 = Static pressure at inlet and exit
- b. The outer diameter of the impeller of a centrifugal pump is 40cm and width of the impeller at outlet is 5cm. The pump is running at 800rpm and is working against a total head of 15m. the vane angle at outlet is 40° and manometric efficiency is 75%. Determine :
 i) Velocity of flow at outlet
 ii) Velocity of water leaving the vane
 iii) Angle made by the absolute velocity at outlet with the direction of motion at outlet
 iv) Discharge (08 Marks)
- c. Explain the phenomenon of surging in compressor. (04 Marks)

OR

- 10 a. Define the following for a centrifugal compressor
 i) Slip and slip coefficient
 ii) Energy transfer
 iii) Power input factor
 iv) Overall pressure ratio
 v) Loading coefficient (10 Marks)
- b. A 4 –stage centrifugal pump has impellers each of 38cms diameter and 1.9cms wide at outlet. The outlet vane angle is 49° and vanes occupy 8% of the outlet area. The manometric efficiency is 84% and overall efficiency is 75%. Determine the head generated by the pump when running at 900rpm discharging 59 litres/second. Also determine the power required. (10 Marks)

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18ME55

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Fluid Power Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a block diagram, explain hydraulic system. (07 Marks)
- b. Give the differences between hydraulic system and pneumatic system. (07 Marks)
- c. Explain Pascal's law. (06 Marks)

OR

- 2 a. With the help of sketch explain filter position in a hydraulic system. (07 Marks)
- b. With a neat sketch, explain water cooled heat exchanger. (07 Marks)
- c. Write a note on Seals. (06 Marks)

Module-2

- 3 a. With a neat sketch, explain internal gear pump. (07 Marks)
- b. A pump having a displacement volume of 90cm^3 delivers $0.082\text{m}^3/\text{min}$ at 1000rpm and 6.9MPa. If the input torque is 102Nm. Find
 - i) Overall efficiency of the pump
 - ii) Theoretical torque required to operate the pump (07 Marks)
- c. With a neat sketch, explain diaphragm type gas loaded accumulator. (06 Marks)

OR

- 4 a. With a neat sketch, explain hydraulic cylinder cushioning. (07 Marks)
- b. A hydraulic motor has a 100cm^3 volumetric displacement. If it has a pressure rating of 140 bars receives oil from a $0.001\text{m}^3/\text{s}$ theoretical flow rate pump, find motor
 - i) Speed
 - ii) Theoretical torque
 - iii) Theoretical power (08 Marks)
- c. With a neat sketch, explain rotary actuator. (05 Marks)

Module-3

- 5 a. With a sketch, explain 3 position 4 way direction control valve. (08 Marks)
- b. Explain working of unloading valve (07 Marks)
- c. Explain working of shuttle valve. (05 Marks)

OR

- 6 a. With the help of circuit diagram, explain sequencing of cylinder. (08 Marks)
- b. Explain metering in and metering out circuits. (12 Marks)

Module-4

- 7 a. List the advantages, disadvantages and applications of Pneumatic system. (08 Marks)
b. With a neat sketch, explain F.R.L unit in a pneumatic system. (12 Marks)

OR

- 8 a. With a neat labelled sketch explain parts of pneumatic double acting cylinder. (07 Marks)
b. With a neat sketch, explain quick exhaust valve. (07 Marks)
c. Explain working of reciprocating air compressor. (06 Marks)

Module-5

- 9 a. With circuit diagram, explain indirect control of single acting cylinders. (08 Marks)
b. Explain 'OR' and 'AND' logic gates. (08 Marks)
c. Write a note on pneumatic throttle valve. (04 Marks)

OR

- 10 a. Explain with circuit coordinated cylinder movements. (10 Marks)
b. With a neat sketch, explain solenoid controlled direction control valve. Mention advantages. (10 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023 Operations Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Operation Management. Explain in brief the functions of operations managements. (10 Marks)
- b. Define Productivity. Explain the factors affecting productivity. (10 Marks)

OR

- 2 a. A glass firm developing a substantial back log of orders is considering three courses of action
 - i) Arrange for sub contracting
 - ii) Begin overtime production construct new facilities

The correct choice depends largely on future demand, which may be low, medium (or) high. By consensus, management ranks the respective probabilities as 0.10, 0.50 and 0.40. A cost analysis reveals the effect on profits as shown below :

Course of action	Profit (in thousand R) if the demand is		
	Low (P = 0.1)	Medium (P = 0.5)	High (P = 0.4)
A. Arrange for sub-constructing	10	50	50
B-Begin over time	-20	60	100
C-construct new facilities	-150	20	200

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- b. Explain break-even analysis with necessary equations, graph and assumptions. (08 Marks)

Module-2

- 3 a. A company adopts method of least squares to develop a linear trend equation for the data as shown in the table below :

Year (X)	1	2	3	4	5	6	7	8	9	10	11
Shipment in tons (Y)	2	3	5	10	8	7	12	14	14	18	19

-
- b. Calculate the trend forecast for the year 12 and 20. (12 Marks)
- c. Explain the following forecasting methods :
 - i) Exponential smoothing
 - ii) Linear Regression (08 Marks)

OR

- 4 a. What is forecasting? List the steps involved in forecasting process. (10 Marks)
- b. A firm use simple exponential smoothing with $\alpha = 0.1$ to forecast demand. The forecast for the week of February 1 was 500 units, where as actual demand turned out to be 450 units.
 - i) Forecast the demand for the week of February 8
 - ii) Assume that the actual demand during the week of the February 8 turned out to be 505units. Forecast the demand for the week of February 15. Continue on forecasting through March 15, assuming the sub sequent demands were actually 516, 488, 467, 554 and 510 units. (10 Marks)

Module-3

- 5 a. Explain the various factors that influence the location of plants. (10 Marks)
 b. Define the following :
 i) Design capacity
 ii) System capacity
 iii) Capacity planning
 iv) Facility layout (10 Marks)

OR

- 6 a. Sketch and explain any two types of layouts. (10 Marks)
 b. What is facility layout? What factors determines the types of layout used in an organization. (10 Marks)

Module-4

- 7 a. Define aggregate planning and master scheduling. Explain the pure strategies used for aggregate planning in brief. (10 Marks)
 b. List the common strategies used in aggregate planning. Explain any two. (10 Marks)

OR

- 8 a. What are the objectives and importance of aggregate planning? (10 Marks)
 b. Briefly explain the following with the help of a flow chart.
 i) Aggregate planning
 ii) Master scheduling (10 Marks)

Module-5

- 9 a. What are the benefits and limitation of MRP? (10 Marks)
 b. State the importance of purchasing and supply management. (10 Marks)

OR

- 10 a. Briefly explain the following :
 i) Vendor Development
 ii) E-procurement
 iii) Concept of tenders
 iv) The procurement process (10 Marks)
 b. Write a note on make or buy decision. (10 Marks)

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Question Paper Version : A

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Which of the following is a possible producer in an ecosystem?
a) Animal b) Plants c) Human beings d) Fish
 2. The largest reservoir of nitrogen in our planet is
a) Oceans b) Biosphere c) Atmosphere d) Rivers
 3. India has the world's largest share of which of the following
a) Manganese b) Mica c) Copper d) Diamond
 4. Identify the non renewable source of energy from the following:
a) Coal b) Fuel cells c) Wind power d) Wave power
 5. Which of the following terminologies is not associated with the vertical structure of forest?
a) Canopy b) Understory c) Forest floor d) First floor
 6. Which of the following is cause of loss of biodiversity?
a) Habitat degradation b) Invasion of non-native species
c) Pollution d) All of these
 7. Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) Cyclone separator
c) Wet collector d) Catalytic converter
 8. When the solid waste consists of large amount of organic matter and if the moisture content is high, which of the following methods of treatment will be ideal?
a) Composting b) Palletizing
c) Incineration d) Recycling

9. Chernobyl Nuclear Disaster occurred in the year
a) 1984 b) 1985 c) 1986 d) 1987
10. The primary cause of acid rain around the world is
a) Carbon dioxide b) Sulphur dioxide
c) Carbon monoxide d) Ozone
11. World Environmental day is held every year on
a) June 5th b) October 2nd c) April 22nd d) November 1st
12. Ozone layer thickness is measured in _____.
a) mm b) cm c) Dobson unit d) Db
13. First of the major environmental protection acts to be promulgated in India was
a) The Water Act b) The Air Act
c) The Environment Act d) Noise Pollution Rules
14. Blue baby syndrome is caused due to _____.
a) Manganese b) Ozone c) Silver d) Nitrate
15. World Earth's day is annually celebrated on
a) April 22nd b) June 5th c) January 1st d) May 1st
16. The most important fuel used by nuclear power plant is
a) U-235 b) U-238 c) U-245 d) U-248
17. Which of the following is a biotic component of ecosystem?
a) Fungi b) Solar light
c) Temperature d) Humidity
18. Abiotic component includes
a) Soil b) Temperature
c) Water d) All of these
19. The word "Environment" is derived from
a) Greek b) French
c) Spanish d) English
20. Which of the following is absorbed by green plants from the atmosphere?
a) Carbon dioxide b) Water
c) Nutrients d) All of these
21. Eutrophication is
a) an improved quality of water in lakes
b) a process in carbon cycle
c) the result to accumulation of plant nutrients in water bodies
d) a water purification technique
22. Primary consumer is
a) Herbivores b) Carnivores
c) Macro consumers d) Omnivores

23. Which among the following is a climatic factor?
a) pressure
b) humidity
c) temperature
d) all of these
24. Biodiversity can be broadly classified into how many types?
a) 2
b) 5
c) 3
d) 4
25. Hot spot areas have
a) Low density of biodiversity
b) Only endangered plants
c) High density of hot springs
d) High density of biodiversity
26. About _____% of the earth's surface is covered by water.
a) 53%
b) 19%
c) 71%
d) 90%
27. Deforestation means
a) preservation of forests
b) destruction of forests
c) monocrop cultivation
d) agriculture
28. When did National Disaster Management Authority formed?
a) 2000
b) 2005
c) 2010
d) 2015
29. Disaster is an event arising out of
a) result of hazard event
b) causes of hazard event
c) causes of disaster event
d) all of these
30. The scientific study of earthquake is called
a) seismograph
b) seismology
c) both a and b
d) none of these
31. South Africa is leading exporter of which mineral?
a) Copper
b) Diamond
c) Silver
d) Gold
32. The word 'sustainable development' came into existence in the year.
a) 1992
b) 1978
c) 1980
d) 1987
33. The other word of landscaping is
a) Reduction
b) Restoration
c) Removing topsoil
d) Restore
34. Cloud seeding with silver iodide is based on the
a) Bergeron process
b) Collision-coalescence process
c) Both a and b
d) None of these
35. Environmental pollution is due to
a) Rapid urbanization
b) Deforestation
c) Afforestation
d) a and b
36. The liquid waste from bathroom and kitchen is called
a) Sullage
b) Domestic sewage
c) Storm water
d) Runoff

37. BOD means
 a) Biochemical Oxygen Demand
 b) Chemical oxygen demand
 c) Biophysical Oxygen Demand
 d) All of these
38. Which of the following source is surface water?
 a) Springs
 b) Streams
 c) Deep wells
 d) All of these
39. Which of the following is biodegradable?
 a) Plastics
 b) Domestic sewage
 c) Detergents
 d) a and c
40. Blaring sounds known to cause
 a) Mental distress
 b) High cholesterol
 c) Neurological problems
 d) All of these
41. "Minamata Disease" is caused due to
 a) Lead
 b) Arsenic
 c) Mercury
 d) Cadmium
42. Alternative eco-friendly fuel for automobiles is
 a) Petrol
 b) Diesel
 c) CNG
 d) Kerosene
43. Population explosion will cause
 a) Biodiversity
 b) Stress on ecosystem
 c) More employment
 d) None of these
44. Which of the following is having high population density?
 a) India
 b) China
 c) USA
 d) Western Europe
45. Demography is the study of
 a) Animals behaviour
 b) Population growth
 c) River
 d) None of these
46. Forest are called as _____.
 a) Air purifier
 b) Earth's lungs
 c) Oxygen reservoir
 d) CO₂ absorbers
47. Which of the following is the facility that the urban people enjoy?
 a) Better quality of air
 b) Better communication access
 c) Large land at cheap rates
 d) None of these
48. Which of the following is an air pollutant?
 a) Carbon dioxide
 b) Oxygen
 c) Nitrogen
 d) Particulate matter
49. Cyoto toxic and expired drugs are disposed of by
 a) damping
 b) autoclave
 c) incineration
 d) chemical disinfection
50. The colour code of plastic bag for disposing of microbial laboratory culture waste.
 a) Black
 b) Red
 c) Blue
 d) White

51. Which of the following element make e-waste hazardous in nature?
a) Land
b) Glass
c) Plastic
d) Iron
52. What is the hazardous pollutant released from batteries?
a) Arsenic
b) Barium
c) Cobalt
d) Cadmium
53. What is the term used for reuse of sewage sludge?
a) Compost
b) Solids
c) Biosolids
d) Sludge
54. Reduction in brightness of the famous Taj Mahal is due to
a) Global warming
b) Air pollution
c) Ozone depletion
d) Afforestation
55. E.I.A. can be expanded as
a) Environment and Industrial Act
b) Environment of Impact Activities
c) Environmental Impact Assessment
d) Environmentally Important Activity
56. Organic Farming is
a) Farming without using pesticides and chemical fertilizer
b) Enhances biodiversity
c) Promotes soil biological activity
d) All of these
57. Bio-remediation means the removal of contaminants from
a) Soil
b) Wastewater
c) Groundwater
d) Both soil and ground water
58. Plants use _____ gas for photosynthesis.
a) Oxygen
b) Methane
c) Nitrogen
d) Carbon dioxide
59. What is the maximum allowable concentration of fluorides in drinking water?
a) 1.0 mg/l
b) 1.25 mg/l
c) 1.50 mg/l
d) 1.75 mg/l
60. Forest rich area in Karnataka is found in _____.
a) Western Ghats
b) Bandipur
c) Nagarhole
d) Mangalore
61. Among the fresh water available in the earth the percentage of surface water is about
a) 50%
b) 10%
c) 5%
d) less than 1%
62. Hepatitis is caused by
a) Protozoa
b) Virus
c) Bacteria
d) Fungus

63. In India groundwater resources are rich in
 a) Plains of river Kaveri and Krishna b) The Deccan plateau
 c) The Gangetic plains d) The plains of Netravati and Kapila
64. The required iron content in drinking water as specified by BIS is
 a) 300 mg/l b) 30 mg/l c) 3 mg/l d) 0.30 mg/l
65. Molasses from sugar industry is used to generate
 a) Biodiesel b) Hydrogen
 c) Bioethanol d) Biomethanol
66. Wind Farms are located in
 a) River basin b) Plain area
 c) Hilly area d) Valley area
67. Biomass consists of
 a) Lignin b) Hemi cellulose
 c) Cellulose d) All of these
68. Natural gas contains
 a) Carbon dioxide b) Hydrogen
 c) Methane d) Nitrogen
69. Anti tobacco day is mentioned on
 a) 31st May b) 30st June
 c) 31st July d) 31st August
70. Population explosion will cause
 a) Socio-Economic Problems b) Food Scarcity
 c) Energy crises d) All of these
71. In geosynchronous orbit altitude of the satellite is about
 a) 36,000 kms b) 10,000 kms
 c) 50,000 kms d) None of these
72. The Air (Prevention and Control of Pollution) Act was enacted in the year.
 a) 1987 b) 1981 c) 1991 d) 1988
73. Kudremukh Iron ore mine, Karnataka was closed due to
 a) River pollution and threat to biodiversity
 b) Land encroachment
 c) Radioactive hazards
 d) Serious health hazard
74. On the eve of Gandhi Jayanthi which andolan was launched by our Honorable Prime Minister
 a) Swadeshi b) Sarvashikshana Abhiyana
 c) Suvarnagrama d) Swach Bharath
75. An international agreement signed in the year 1987, to protect stratospheric ozone is known as
 a) Montreal protocol b) Kyoto protocol
 c) Earth summit d) None of these

76. The explosion of First Atomic Bomb was done in Hiroshima and Nagasaki in
 a) 1946 b) 1986 c) 1945 d) 1947
77. A dangerous pesticide which has been reported to cause physical deformities to people of Kerala and Karnataka states
 a) Endosulfan b) Fluorides c) DDT d) Dioxygene
78. Visible portion of EMR ranges between
 a) 0.4 - 0.76 μm b) 10.5 - 12.5 μm c) 8.0 - 14.0 μm d) None of these
79. Data representation in Raster data is by
 a) pixel b) points, lines and polygon
 c) latitude and longitude d) none of these
80. In water treatment, alum is used for
 a) softening b) coagulation c) filtration d) disinfection
81. GIS stands for
 a) Geostationary Interact Sector b) Geographical Information System
 c) Geotechnical Information Society d) Geothermal Investigation Site
82. LPG is a mixture of
 a) N_2 and H_2S b) CO_2 and N_2
 c) Propane and butanes d) Methane and CO_2
83. The Tiger Conservation Project was started in
 a) 1973 b) 1975 c) 1981 d) 2000
84. The leader of "Chipko Movement" is
 a) Sunderlal Bahuguna b) Medha Patkar
 c) Vandana Shiva d) Mahatma Gandhi
85. Which of the following is the source of Fly-ash?
 a) Vehicular exhaust b) Sewage
 c) Thermal power plant d) All of these
86. The permissible range of pH for drinking water as per the Indian Standard is
 a) 6 to 9 b) 6.5 to 8.5
 c) 6 to 8.5 d) 6.5 to 7.5
87. Water logging is a phenomena in which
 a) Water patterns are rotated
 b) Soil root zone becomes saturated due to over irrigation
 c) Erosion of soil
 d) Soil degradation
88. Carbon content is higher in
 a) Living matter b) Soil
 c) Water d) Atmosphere
89. Springs means
 a) Surface water b) Atmosphere water
 c) Both (a) and (b) d) Ground water

90. Bio-remediation means deliberately introducing micro organisms to break pollutants.
a) Soil b) Waste water
c) Ground water d) Both soil and groundwater
91. The Karnataka State Pollution Control Board (KSPCB) was established in the year.
a) 1974 b) 1982 c) 1973 d) 1983
92. Which of the following is not a part of the hydrological cycle?
a) Precipitation b) Infiltration
c) Transpiration d) Perspiration
93. First International Earth Summit was hold at
a) USA b) Russia
c) Rio-de-Janerio d) Johannesburg
94. Which among the following has highest percentage of calorific value?
a) Anthracite b) Peat
c) Lignite d) Bituminous coal
95. Nitrogen fixing bacteria exists in
a) Leaf b) Stem
c) Roots d) Flower
96. The two major components of ecosystem are
a) Adiabatic and isotropic b) Ecologic and climatologic
c) Cyclic and biologic d) Abiotic and biotic
97. Geothermal energy is a
a) Heat energy b) Wind energy
c) Current energy d) Solar energy
98. The average life expectancy around the world is currently.
a) Decreasing b) Increasing
c) Stabilizing d) Not changing
99. The universal declaration of Human Rights was proclaimed by the UN in the year.
a) 1946 b) 1947
c) 1948 d) 1949
100. The objective of Integrated Child Development Service (ICDS) are
a) Immunization b) Health check up and referral services
c) Pre-school non formal education d) All of these

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CBCS SCHEME

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Question Paper Version : B

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies

(COMMON TO ALL BRANCHES)

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

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- [illegible]

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73. Which among the following is a climatic factor?
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c) temperature d) all of these
74. Biodiversity can be broadly classified into how many types?
a) 2 b) 5 c) 3 d) 4

75. Hot spot areas have
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 b) Only endangered plants
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89. The word "Environment" is derived from
 a) Greek
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- the following is absorbed by green plants from the atmosphere?
a) Carbon dioxide
b) Water
c) Nitrogen
d) All of these
- the following is a possible producer in an ecosystem?
a) Fungi
b) Plants
c) Human beings
d) All of these
- the largest reservoir of nitrogen in our planet is
a) Hydrosphere
b) Biosphere
c) Atmosphere
d) Lithosphere
- the world's largest share of which of the following is in India?
a) Iron
b) Mica
c) Copper
d) Coal
- the non renewable source of energy from the following:
a) Solar energy
b) Fuel cells
c) Wind power
d) Biomass
- the following terminologies is not associated with the forest?
a) Canopy
b) Understory
c) Forest floor
d) All of these
- the following is cause of loss of biodiversity?
a) Deforestation
b) Invasion of non-native species
c) Pollution
d) All of these
- the pollution from automobiles can be controlled by fitting
a) Catalytic converter
b) Cyclone separator
c) Air filter
d) Catalytic converter
- the solid waste consists of large amount of organic matter, which is highly biodegradable, which of the following methods of treatment will be most suitable?
a) Incineration
b) Palletizing
c) Landfilling
d) Recycling
- the Bhopal Nuclear Disaster occurred in the year
a) 1984
b) 1985
c) 1986
d) 1987
- the primary cause of acid rain around the world is
a) Carbon dioxide
b) Sulphur dioxide
c) Nitrogen dioxide
d) Ozone
- * * * * *

the following is absorbed by green plants from the atmosphere?
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c) Electrostatic precipitator
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d) Recycling

the Bhopal Nuclear Disaster occurred in the year
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c) 1986
d) 1987

the primary cause of acid rain around the world is
a) Carbon dioxide
b) Sulphur dioxide
c) Nitrogen monoxide
d) Ozone

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Question Paper Version : C

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
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 - b) Virus
 - c) Bacteria
 - d) Fungus
13. In India groundwater resources are rich in
 - a) Plains of river Kaveri and Krishna
 - b) The Deccan plateau
 - c) The Gangetic plains
 - d) The plains of Netravati and Kapila
14. The required iron content in drinking water as specified by BIS is
 - a) 300 mg/l
 - b) 30 mg/l
 - c) 3 mg/l
 - d) 0.30 mg/l
15. Molasses from sugar industry is used to generate
 - a) Biodiesel
 - b) Hydrogen
 - c) Bioethanol
 - d) Biomethanol
16. Wind Farms are located in
 - a) River basin
 - b) Plain area
 - c) Hilly area
 - d) Valley area
17. Biomass consists of
 - a) Lignin
 - b) Hemi cellulose
 - c) Cellulose
 - d) All of these
18. Natural gas contains
 - a) Carbon dioxide
 - b) Hydrogen
 - c) Methane
 - d) Nitrogen
19. Anti tobacco day is mentioned on
 - a) 31st May
 - b) 30st June
 - c) 31st July
 - d) 31st August
20. Population explosion will cause
 - a) Socio-Economic Problems
 - b) Food Scarcity
 - c) Energy crises
 - d) All of these
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 - a) Geostationary Interact Sector
 - b) Geographical Information System
 - c) Geotechnical Information Society
 - d) Geothermal Investigation Site
22. LPG is a mixture of
 - a) N₂ and H₂S
 - b) CO₂ and N₂
 - c) Propane and butanes
 - d) Methane and CO₂

23. The Tiger Conservation Project was started in
 a) 1973 b) 1975 c) 1981 d) 2000
24. The leader of “Chipko Movement” is
 a) Sunderlal Bahuguna b) Medha Patkar
 c) Vandana Shiva d) Mahatma Gandhi
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 a) 6 to 9 b) 6.5 to 8.5
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 a) Water patterns are rotated
 b) Soil root zone becomes saturated due to over irrigation
 c) Erosion of soil
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28. Carbon content is higher in
 a) Living matter b) Soil
 c) Water d) Atmosphere
29. Springs means
 a) Surface water b) Atmosphere water
 c) Both (a) and (b) d) Ground water
30. Bio-remediation means deliberately introducing micro organisms to break pollutants.
 a) Soil b) Waste water
 c) Ground water d) Both soil and groundwater
31. Which of the following is a possible producer in an ecosystem?
 a) Animal b) Plants c) Human beings d) Fish
32. The largest reservoir of nitrogen in our planet is
 a) Oceans b) Biosphere c) Atmosphere d) Rivers
33. India has the world’s largest share of which of the following
 a) Manganese b) Mica c) Copper d) Diamond
34. Identify the non renewable source of energy from the following:
 a) Coal b) Fuel cells c) Wind power d) Wave power
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 a) Canopy b) Understory c) Forest floor d) First floor
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 c) Carbon monoxide d) Ozone
41. Eutrophication is
 a) an improved quality of water in lakes
 b) a process in carbon cycle
 c) the result to accumulation of plant nutrients in water bodies
 d) a water purification technique
42. Primary consumer is
 a) Herbivores b) Carnivores
 c) Macro consumers d) Omnivores
43. Which among the following is a climatic factor?
 a) pressure b) humidity
 c) temperature d) all of these
44. Biodiversity can be broadly classified into how many types?
 a) 2 b) 5 c) 3 d) 4
45. Hot spot areas have
 a) Low density of biodiversity b) Only endangered plants
 c) High density of hot springs d) High density of biodiversity
46. About _____% of the earth's surface is covered by water.
 a) 53% b) 19% c) 71% d) 90%
47. Deforestation means
 a) preservation of forests b) destruction of forests
 c) monocrop cultivation d) agriculture
48. When did National Disaster Management Authority formed?
 a) 2000 b) 2005 c) 2010 d) 2015
49. Disaster is an event arising out of
 a) result of hazard event b) causes of hazard event
 c) causes of disaster event d) all of these
50. The scientific study of earthquake is called
 a) seismograph b) seismology
 c) both a and b d) none of these

51. World Environmental day is held every year on
 a) June 5th b) October 2nd c) April 22nd d) November 1st
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 a) mm b) cm c) Dobson unit d) Db
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 a) Fungi b) Solar light
 c) Temperature d) Humidity
58. Abiotic component includes
 a) Soil b) Temperature
 c) Water d) All of these
59. The word "Environment" is derived from
 a) Greek b) French
 c) Spanish d) English
60. Which of the following is absorbed by green plants from the atmosphere?
 a) Carbon dioxide b) Water
 c) Nutrients d) All of these
61. The Karnataka State Pollution Control Board (KSPCB) was established in the year.
 a) 1974 b) 1982 c) 1973 d) 1983
62. Which of the following is not a part of the hydrological cycle?
 a) Precipitation b) Infiltration
 c) Transpiration d) Perspiration
63. First International Earth Summit was held at
 a) USA b) Russia
 c) Rio-de-Janeiro d) Johannesburg
64. Which among the following has highest percentage of calorific value?
 a) Anthracite b) Peat
 c) Lignite d) Bituminous coal
65. Nitrogen fixing bacteria exists in
 a) Leaf b) Stem
 c) Roots d) Flower

66. The two major components of ecosystem are
a) Adiabatic and isotropic
b) Ecologic and climatologic
c) Cyclic and biologic
d) Abiotic and biotic
67. Geothermal energy is a
a) Heat energy
b) Wind energy
c) Current energy
d) Solar energy
68. The average life expectancy around the world is currently.
a) Decreasing
b) Increasing
c) Stabilizing
d) Not changing
69. The universal declaration of Human Rights was proclaimed by the UN in the year.
a) 1946
b) 1947
c) 1948
d) 1949
70. The objective of Integrated Child Development Service (ICDS) are
a) Immunization
b) Health check up and referral services
c) Pre-school non formal education
d) All of these
71. South Africa is leading exporter of which mineral?
a) Copper
b) Diamond
c) Silver
d) Gold
72. The word 'sustainable development' came into existence in the year.
a) 1992
b) 1978
c) 1980
d) 1987
73. The other word of landscaping is
a) Reduction
b) Restoration
c) Removing topsoil
d) Restore
74. Cloud seeding with silver iodide is based on the
a) Bergeron process
b) Collision-coalescence process
c) Both a and b
d) None of these
75. Environmental pollution is due to
a) Rapid urbanization
b) Deforestation
c) Afforestation
d) a and b
76. The liquid waste from bathroom and kitchen is called
a) Sullage
b) Domestic sewage
c) Storm water
d) Runoff
77. BOD means
a) Biochemical Oxygen Demand
b) Chemical oxygen demand
c) Biophysical Oxygen Demand
d) All of these
78. Which of the following source is surface water?
a) Springs
b) Streams
c) Deep wells
d) All of these
79. Which of the following is biodegradable?
a) Plastics
b) Domestic sewage
c) Detergents
d) a and c

80. Blaring sounds known to cause
 a) Mental distress
 b) High cholesterol
 c) Neurological problems
 d) All of these
81. In geosynchronous orbit altitude of the satellite is about
 a) 36,000 kms
 b) 10,000 kms
 c) 50,000 kms
 d) None of these
82. The Air (Prevention and Control of Pollution) Act was enacted in the year.
 a) 1987
 b) 1981
 c) 1991
 d) 1988
83. Kudremukh Iron ore mine, Karnataka was closed due to
 a) River pollution and threat to biodiversity
 b) Land encroachment
 c) Radioactive hazards
 d) Serious health hazard
84. On the eve of Gandhi Jayanthi which andolan was launched by our Honorable Prime Minister
 a) Swadeshi
 b) Sarvashikshana Abhiyana
 c) Suvarnagrama
 d) Swach Bharath
85. An international agreement signed in the year 1987, to protect stratospheric ozone is known as
 a) Montreal protocol
 b) Kyoto protocol
 c) Earth summit
 d) None of these
86. The explosion of First Atomic Bomb was done in Hiroshima and Nagasaki in
 a) 1946
 b) 1986
 c) 1945
 d) 1947
87. A dangerous pesticide which has been reported to cause physical deformities to people of Kerala and Karnataka states
 a) Endosulfan
 b) Fluorides
 c) DDT
 d) Dioxygene
88. Visible portion of EMR ranges between
 a) 0.4 - 0.76 μm
 b) 10.5 - 12.5 μm
 c) 8.0 - 14.0 μm
 d) None of these
89. Data representation in Raster data is by
 a) pixel
 b) points, lines and polygon
 c) latitude and longitude
 d) none of these
90. In water treatment, alum is used for
 a) softening
 b) coagulation
 c) filtration
 d) disinfection
91. Which of the following element make e-waste hazardous in nature?
 a) Land
 b) Glass
 c) Plastic
 d) Iron
92. What is the hazardous pollutant released from batteries?
 a) Arsenic
 b) Barium
 c) Cobalt
 d) Cadmium
93. What is the term used for reuse of sewage sludge?
 a) Compost
 b) Solids
 c) Biosolids
 d) Sludge

94. Reduction in brightness of the famous Taj Mahal is due to
a) Global warming
b) Air pollution
c) Ozone depletion
d) Afforestation
95. E.I.A. can be expanded as
a) Environment and Industrial Act
b) Environment of Impact Activities
c) Environmental Impact Assessment
d) Environmentally Important Activity
96. Organic Farming is
a) Farming without using pesticides and chemical fertilizer
b) Enhances biodiversity
c) Promotes soil biological activity
d) All of these
97. Bio-remediation means the removal of contaminants from
a) Soil
b) Wastewater
c) Groundwater
d) Both soil and ground water
98. Plants use _____ gas for photosynthesis.
a) Oxygen
b) Methane
c) Nitrogen
d) Carbon dioxide
99. What is the maximum allowable concentration of fluorides in drinking water?
a) 1.0 mg/l
b) 1.25 mg/l
c) 1.50 mg/l
d) 1.75 mg/l
100. Forest rich area in Karnataka is found in _____.
a) Western Ghats
b) Bandipur
c) Nagarhole
d) Mangalore

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Question Paper Version : D

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. World Environmental day is held every year on
a) June 5th b) October 2nd c) April 22nd d) November 1st
 2. Ozone layer thickness is measured in _____.
a) mm b) cm c) Dobson unit d) Db
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a) The Water Act b) The Air Act
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a) U-235 b) U-238 c) U-245 d) U-248
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 8. Abiotic component includes
a) Soil b) Temperature
c) Water d) All of these

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 - a) Greek
 - b) French
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 - b) Barium
 - c) Cobalt
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- The term used for reuse of sewage sludge?
 a) Solids
 b) Solids
 c) Sludge
 d) Sludge
- The loss in brightness of the famous Taj Mahal is due to
 a) Global warming
 b) Air pollution
 c) Acid depletion
 d) Afforestation
- The Environment Protection Act can be expanded as
 a) Environment Protection and Industrial Act
 b) Environment of India
 c) Environmental Impact Assessment
 d) Environmentally Impact Assessment
- Organic farming is
 a) Farming without using pesticides and chemical fertilizer
 b) Farming without using pesticides and chemical fertilizer
 c) Farming without using pesticides and chemical fertilizer
 d) Farming without using pesticides and chemical fertilizer
- Organic farming
 a) Increases biodiversity
 b) Increases biodiversity
 c) Increases biodiversity
 d) Increases biodiversity
- Organic farming
 a) Increases soil biological activity
 b) Increases soil biological activity
 c) Increases soil biological activity
 d) Increases soil biological activity
- Organic farming
 a) Increases these
 b) Increases these
 c) Increases these
 d) Increases these
- Organic farming
 a) Organic farming means the removal of contaminants from
 b) Organic farming means the removal of contaminants from
 c) Organic farming means the removal of contaminants from
 d) Organic farming means the removal of contaminants from
- Organic farming
 a) Wastewater
 b) Wastewater
 c) Wastewater
 d) Wastewater
- Organic farming
 a) Both soil and groundwater
 b) Both soil and groundwater
 c) Both soil and groundwater
 d) Both soil and groundwater
- Organic farming
 a) Carbon dioxide gas for photosynthesis.
 b) Carbon dioxide gas for photosynthesis.
 c) Carbon dioxide gas for photosynthesis.
 d) Carbon dioxide gas for photosynthesis.
- Organic farming
 a) Methane
 b) Methane
 c) Methane
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- Organic farming
 a) Carbon dioxide
 b) Carbon dioxide
 c) Carbon dioxide
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- Organic farming
 a) The maximum allowable concentration of fluorides in drinking water is 1.5 mg/l
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- Organic farming
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 a) The synchronous orbit altitude of the satellite is about
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- Organic farming
 a) 10,000 kms
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- Organic farming
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 a) The Prevention and Control of Pollution) Act was enacted in 1986
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 - a) 1946
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 - c) Atmosphere
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- ong the following is a climatic factor?
a) temperature
b) humidity
c) wind
d) all of these
- World climate can be broadly classified into how many types?
a) 2
b) 5
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- Which of the following areas have high biodiversity?
a) Tropic of Cancer
b) Only endangered plants
c) Tropic of Capricorn
d) High density of biotic communities
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a) 71%
b) 19%
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d) all of these
- The scientific study of earthquake is called
a) seismology
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c) seismograph
d) none of these
- "Mad Hatter's Disease" is caused due to
a) Lead
b) Arsenic
c) Mercury
d) Cadmium
- _____ is an eco-friendly fuel for automobiles is
a) CNG
b) Diesel
c) CNG
d) Petrol
- _____ explosion will cause global warming
a) Nuclear
b) Stress on ecosystem
c) Greenhouse
d) None of these
- Which of the following is having high population density?
a) China
b) India
c) USA
d) Brazil

78. Which of the following is an air pollutant?
 a) Carbon dioxide
 b) Oxygen
 c) Nitrogen
 d) Particulate matter
79. Cyto toxic and expired drugs are disposed of by
 a) damping
 b) autoclave
 c) incineration
 d) chemical disinfection
80. The colour code of plastic bag for disposing of microbial laboratory culture waste.
 a) Black
 b) Red
 c) Blue
 d) White
81. Among the fresh water available in the earth the percentage of surface water is about
 a) 50%
 b) 10%
 c) 5%
 d) less than 1%
82. Hepatitis is caused by
 a) Protozoa
 b) Virus
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83. In India groundwater resources are rich in
 a) Plains of river Kaveri and Krishna
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 c) Geotechnical Information Society
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92. LPG is a mixture of
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a) Soil b) Waste water
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* * * * *

CBCS SCHEME

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18ME71

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Control Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a block diagram, explain automobile speed closed control system. (10 Marks)
- b. List and explain requirements of an ideal control system. (10 Marks)

OR

- 2 a. Explain: (i) Proportional controller (ii) Derivative controller (10 Marks)
- b. Obtain transfer function for armature controlled D-C motor. (10 Marks)

Module-2

- 3 a. Explain typical test signals in control system. (10 Marks)
- b. Determine order and type for open and closed loop control system as shown in Fig.Q3(b).

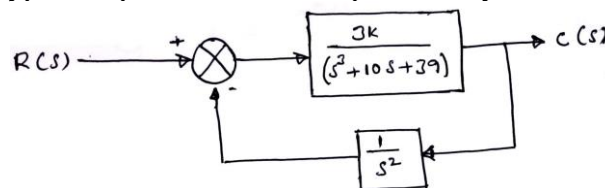


Fig.Q3(b)

(10 Marks)

OR

- 4 a. Define:

(i) Delay time
(iv) Maximum overshoot

(ii) Rise time
(v) Setting time

(iii) Peak time
(vi) Peak time
- b. A unity feedback system is characterized by open loop transfer function $G(s) = \frac{16}{s^2 + 2s + 16}$. Determine the following when the system subjected to unit step input :

(i) Undamped not usual frequency
(iv) Peak time

(ii) Damping ratio
(v) Settling time

(iii) Peak overshoot
(vi) Peak time

(10 Marks)

(10 Marks)

Module-3

- 5 a. Reduce the block diagram as shown in Fig.Q5(a) to simple form and find transfer function:

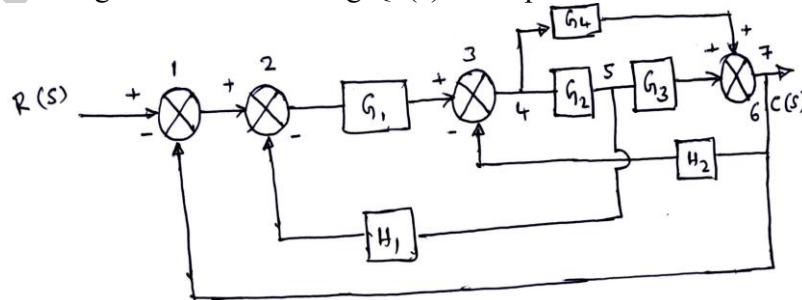


Fig.Q5(a)

(10 Marks)

- b. Obtain transfer function of block diagram shown in Fig.Q5(b) by reduction technique.

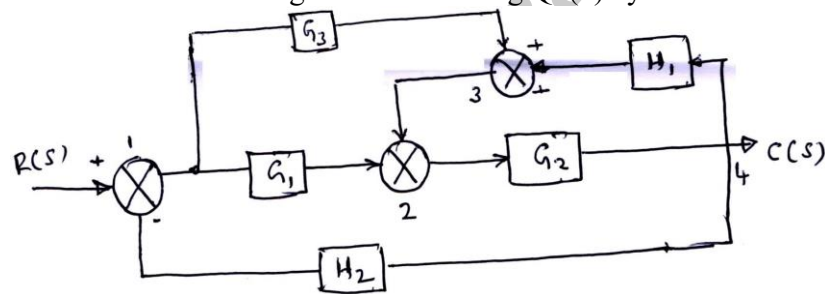


Fig.Q5(b)

(10 Marks)

OR

- 6 a. For the system shown in Fig.Q6(a), determine $\frac{C(s)}{R(s)}$ using Mason's gain formula.

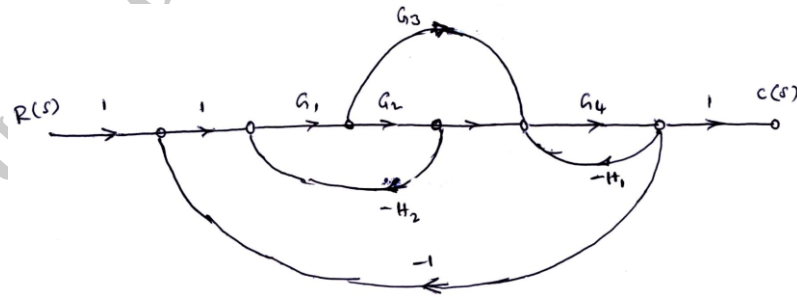


Fig.Q6(a)

(10 Marks)

- b. Using SFG and Mason's gain formula, obtain the overall transfer function of system shown in Fig.Q6(b).

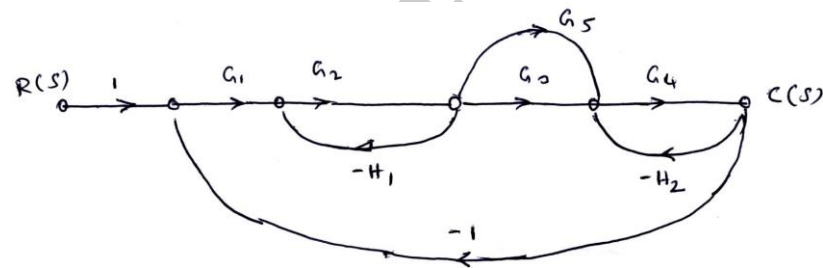


Fig.Q6(b)

(10 Marks)

Module-4

- 7 a. Applying Routh criterion, discuss the stability of closed loop system as function for open loop transfer function:

$$G(s)H(s) = \frac{K(s+1)}{s(s-1)(s^2+4s+16)}$$

(10 Marks)

- b. Investigate the stability of system using Routh Hurwitz criterion having characteristic equation $s^5 + 4s^4 + 12s^3 + 20s^2 + 30s + 100 = 0$

(10 Marks)

OR

- 8 Sketch the root locus for negative feedback system whose open loop transfer function is

$$\text{given by } G(s)H(s) = \frac{K}{s(s+3)(s^2+3s+4.5)}$$

(20 Marks)

Module-5

- 9 a. Sketch polar plot for transfer function $G(s) = \frac{10}{s(s+1)(s+2)}$. (10 Marks)
- b. Open loop function control system $G(s)H(s) = \frac{1}{s^2(s+2)}$, sketch Nyquist plot and ascertain stability. (10 Marks)

OR

- 10 A unity feedback control system has $G(s) = \frac{80}{s(s+2)(s+20)}$. Draw the Bode plot if phase cross over occur at $\omega = 6.35$ rad/sec, find the corresponding gain margin. (20 Marks)

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CBCS SCHEME

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18ME71

Seventh Semester B.E. Degree Examination, June/July 2023

Control Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define control system with block diagram. Explain open loop and closed loop control system. (08 Marks)
- b. Explain the following controllers :
 - i) Proportional controllers
 - ii) Proportional plus integral controller. (06 Marks)
- c. Obtain the differential equation and determine the transfer function of mechanical networks shown in Fig.Q1(c).

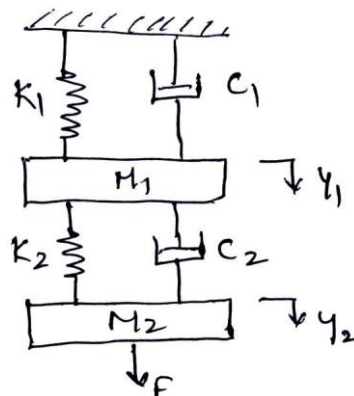


Fig.Q1(c)

(06 Marks)

OR

- 2 a. What are the requirements of a Good Control System? Distinguish between open loop and closed loop control system. (06 Marks)
- b. Explain the following controllers with block diagram :
 - i) Integral controller
 - ii) Derivative controller
 - iii) Proportional plus integral plus differential controllers. (09 Marks)
- c. Derive an expression for transfer function of hydraulic system. (05 Marks)

Module-2

- 3 a. What are standard test signals? Derive an expression for transient response of first order system subjected to step input I/P. (05 Marks)
- b. Derive an expression for steady state error and explain error constants. (05 Marks)
- c. A unity feedback system is characterized by an open loop transfer function :

$$G(s) = \frac{K}{s(s+10)}$$

Determine the gain K so that, the system will have a damping ratio of 0.5. For this value of K determine the settling time, peak overshoot and time of peak overshoot for unit step input I/P. (10 Marks)

OR

- 4 a. Explain with the help of neat sketch transient response specifications of second order under damped system. (06 Marks)
- b. An underdamped second order system is subjected to a step input of 4 units. If the first peak overshoot of 25% occurs at a time equal to 0.8 seconds. Then determine rise time, settling time, damping co-efficient (factor) and natural frequency. (08 Marks)
- c. A unity feedback system has $G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$. Determine : i) Type of system
ii) All error co-efficient iii) Steady state error for ramp input with magnitude 4. (06 Marks)

Module-3

- 5 a. What is block diagram? Obtain the transfer function $C(s)/R(s)$ for the following Fig.Q5(a) using block diagram reduction rules.

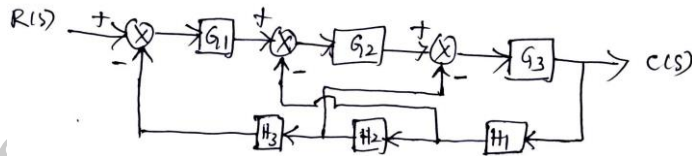


Fig.Q5(a)

(10 Marks)

- b. Define the terms : Node, Branch, Transmittance, Loop gain, Forward path, Source, Non-touching loops and also explain Masson's Gain Formulae. (10 Marks)

OR

- 6 a. Construct the signal flow graph for the following set of system equations and find the transfer function :
 $Y_2 = G_1 Y_1 + G_3 Y_3$; $Y_3 = G_4 Y_1 + G_2 Y_2 + G_5 Y_3$ and $Y_4 = G_6 Y_2 + G_7 Y_3$. (10 Marks)
- b. Draw the signal flow graph for the system shown in Fig.Q6(b) and determine. The transfer function using Masson's gain formulae.

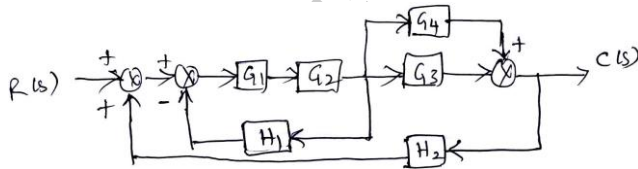


Fig.Q6(b)

(10 Marks)

Module-4

- 7 a. State and explain the Routh's stability criterion. The characteristic equation of a system is given by $s^4 + 6s^3 + 23s^2 + 40s + 50 = 0$. Determine the stability using R-H criterion. (06 Marks)
- b. The characteristics equation of a system is given by $s^4 + 6s^3 + 11s^2 + K = 0$. Determine the range of K for the system to be stable. Use R - H criterion, (06 Marks)
- c. Sketch the root locus plot of a unity feedback system with an open loop transfer function :
 $G(s) = \frac{K}{s(s+2)(s+4)}$. What is the greatest value of K which can be used before continuous oscillations occurs. Also determine the frequency of continuous oscillations. (08 Marks)

OR

- 8 a. Investigate the stability of the system using Routh Henvitz criterion having the following characteristics in $s^5 + 4s^4 + 12s^3 + 20s^2 + 30s + 100 = 0$. (08 Marks)
- b. Sketch the root locus plot for the transfer function : $G(s)H(s) = \frac{K}{s(s^2 + 2s + 2)}$. For what value of K will the system be unstable? Find the frequency at which the locus crosses the imaginary axis. (12 Marks)

Module-5

- 9 a. Explain Nyquist Stability Criterion. (04 Marks)
- b. Sketch the polar plot for $GH(s) = \frac{12}{s(s+2)(s+4)}$ and ascertain the nature of stability. (06 Marks)
- c. Sketch the bode plot and determine the gain crossover and phase crossover frequency, $GH(s) = \frac{10}{s(1+0.55s)(1+0.1s)}$. (10 Marks)

OR

- 10 a. For a system with an open loop transformation, $GH(s) = \frac{1}{s(1+2s)(1+s)}$ Comment on stability of the system by Nyquist plot. (08 Marks)
- b. Draw the bode plot for a system having $G(s)H(s) = \frac{100}{s(s+1)(s+2)}$. Find, Gain Margin, Phase margin, Gain crossover frequency and phase crossover frequency. (12 Marks)

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CBCS SCHEME

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18ME72

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023 Computer Aided Design and Manufacturing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Automation? List different types of automation and discuss with an example. (10 Marks)
- b. Define : (i) Production rate
(ii) Production capacity
(iii) Utilization and availability
(iv) MLT and WIP (10 Marks)

OR

- 2 a. Sketch and explain any two types of Automated flow lines. (10 Marks)
- b. The ideal cycle time of an 16 station transfer line is 1.4 min. The average down time per line will be 6 min and the probability of break downs per cycle is equal for all cycles and is equal to 0.004. Determine production rate and line efficiency by considering both upper bound and lower bound approaches. (10 Marks)

Module-2

- 3 a. Briefly explain design process and the application of computer in design process. (10 Marks)
- b. Explain the following in detail:
Translation , Rotation , Concatenation and benefits of CAD. (10 Marks)

OR

- 4 a. What do you understand by CAPP? With a block diagram explain Generative System. (10 Marks)
- b. Write a note on MRP Inputs and Outputs, Benefits of MRP. (10 Marks)

Module-3

- 5 a. Define Group Technology. List various types of FMS and benefits of FMS. (10 Marks)
- b. What do you mean by As/Rs? Explain briefly about Part Identification System. (10 Marks)

OR

- 6 a. A manual assembly line has to accomplish 10 work elements to complete the assembly. The element times and precedence requirements are listed in the table. The production rate of the line is 60 units per hour. The efficiency of the line is 95% and the repositioning time is 3 sec. Use Kilbridge and Westers method to balance the line and compute balance delay and balance efficiency.

Element	1	2	3	4	5	6	7	8	9	10
T _e (min)	0.3	0.4	0.3	0.2	0.4	0.1	0.5	0.6	0.4	0.6
Preceded by	-	-	1	1, 2	2	3, 4	4	5	6, 7	8, 9

- b. From above data compute balance delay and balance efficiency using RPW method. (10 Marks)

Module-4

- 7 a. Define CNC. Enlist various advantages / disadvantages and application of CNC. (10 Marks)
 b. List few G and M codes you came across and write a program to cut the profile shown in Fig.Q7(b).

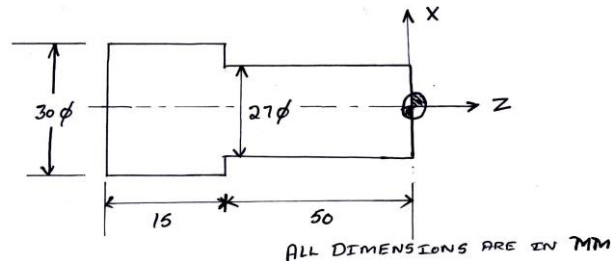


Fig.Q7(b)

(10 Marks)

OR

- 8 a. With neat sketches show robot components and joints. (10 Marks)
 b. List various configuration of a Industrial robot, sketch and draw in detail. (10 Marks)

Module-5

- 9 a. Discuss the basic principles of additive manufacturing and list various advantages / limitations of AM technique. (10 Marks)
 b. Explain the process in brief photopolymerization, material jetting. (10 Marks)

OR

- 10 a. Describe Slicing in AM. (10 Marks)
 b. Explain the following :
 (i) Direct Energy deposition (ii) Sheet lamination. (10 Marks)

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Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Total Quality Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Quality. Explain quality in daily life in middle age during Industrial revolution. (10 Marks)
- b. What are the factors affecting Quality? (10 Marks)

OR

- 2 a. Explain the list of dimension of Quality. (10 Marks)
- b. Explain the contribution of Gurus Quality. (10 Marks)

Module-2

- 3 a. Explain the Modern method of Leadership. (10 Marks)
- b. What are the duties of Quality Control? (10 Marks)

OR

- 4 a. Mention the Deming's Philosophy of 14 points. (10 Marks)
- b. What are the future requirements for the short and long term factors affects in the Organisation? (10 Marks)

Module-3

- 5 a. Explain with neat sketch, the Kano model. (10 Marks)
- b. What are the elements present in structure of Quality circle? (10 Marks)

OR

- 6 a. Mention the Tangible and Intangible benefits through TQM. (10 Marks)
- b. Who was Malcolm Baldrige? What is the Malcolm Baldrige National Quality Award? What was the award established? (10 Marks)

Module-4

- 7 a. What is Six Sigma? Define Six Sigma and phases of Six Sigma. (10 Marks)
- b. With neat analysis of graph, explain Pareto Analysis. (10 Marks)

OR

- 8 a. Define Process of Operation of Quality circle and steps. (10 Marks)
- b. What are the benefits of Forming Quality circles? (10 Marks)

Module-5

- 9 a. What is meant by Total Productive Maintenance, with an example? (10 Marks)
- b. Define Quality by Design in TQM and What are elements of Quality by design? (10 Marks)

OR

- 10 a. Define the Environmental Management Systems and what is the importance. (10 Marks)
- b. Explain EMS under ISO 14001. What is the cost and benefits? (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8 = 50, will be treated as malpractice.

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18ME734

Seventh Semester B.E. Degree Examination, June/July 2023 Total Quality Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Write about any two Quality Guru's contribution to TQM. (10 Marks)
- b. What are the obstacles in TQM implementations? (10 Marks)

OR

- 2 a. What is QMS? Explain different series of ISO 9000 standards. (10 Marks)
- b. Draw and explain the TQM Frame Work. (10 Marks)

Module-2

- 3 a. Explain Deming's view on leadership through his fourteen points. (10 Marks)
- b. Explain the characteristics of Quality Leader. (06 Marks)
- c. Explain different ways of communication. (04 Marks)

OR

- 4 a. Define Ethics. What are the roots causes of unethical behaviour? (10 Marks)
- b. Write note on strategic planning. (06 Marks)
- c. Explain briefly decision making process. (04 Marks)

Module-3

- 5 a. What is customer satisfaction? What are the perceptions of customer about quality? (10 Marks)
- b. Explain Kano-model of conceptualities customer requirements. (10 Marks)

OR

- 6 a. Describe briefly the customer retention. (04 Marks)
- b. Explain the role of employee's involvement in customer satisfaction. (08 Marks)
- c. Describe recognition and reward. (08 Marks)

Module-4

- 7 a. Explain: (i) Process capability (ii) Six-Sigma (10 Marks)
- b. Explain: (i) 5S (ii) Kaizen (10 Marks)

OR

- 8 a. Explain with example cause and effect diagram. (10 Marks)
- b. Discuss the control charts for variable and attribute. (10 Marks)

Module-5

- 9 a. What is total productive maintenance? What are its objectives? (08 Marks)
- b. Define Quality by Design. What are its key components? (08 Marks)
- c. Write note on types of maintenance. (04 Marks)

OR

- 10 a. Define EMS and explain its benefits. (04 Marks)
- b. Explain the basic elements of EMS under ISO14001. (08 Marks)
- c. Describe the benefit of QSD and its implementation. (08 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8=50, will be treated as malpractice.

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Seventh Semester B.E. Degree Examination, Jan./Feb. 2023
Mechatronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Mechatronics. Explain with block diagram, the basic elements of feedback control system. (10 Marks)
- b. What is a sequential controller? Explain with neat block diagram, the working of an automated washing machine. (10 Marks)

OR

- 2 a. Explain the working principle of hall effect sensor. Discuss how a hall effect sensor is used in fluid level detector. (10 Marks)
- b. How does the following work : (i) Photo diode (ii) Photo transistor (iii) Photo resistor. (06 Marks)
- c. Explain how an Eddy current proximity sensor works. (04 Marks)

Module-2

- 3 a. Explain with neat sketch the working of a 741 op amp. (05 Marks)
- b. Explain the signal conditioning process. (07 Marks)
- c. What is a filter? How are filters classified? Write brief note on types of filters. (08 Marks)

OR

- 4 a. What is a relay? Explain different types of electromagnetic relays and its working with circuit symbol. (06 Marks)
- b. With neat block diagram, explain how speed control in DC motors is achieved. (06 Marks)
- c. Explain the working of variable reluctance stepper motor with neat sketch. (08 Marks)

Module-3

- 5 a. Differentiate between microprocessor and a microcontroller. (05 Marks)
- b. Explain briefly the following forms of memory units :
ROM ; PROM ; EPROM ; EEPROM ; RAM (10 Marks)
- c. Explain basic elements of a control system with neat sketch. (05 Marks)

OR

- 6 a. Explain with neat layout the internal architecture of INTEL 8085A microprocessor. (10 Marks)
- b. Write short notes on "BUS" related to 8085 microprocessor. (05 Marks)
- c. How does a program counter work? (05 Marks)

Module-4

- 7 a. Explain in detail the criteria used for selection of a PLC. (10 Marks)
- b. What is a PLC? Explain the basic structure of PLC with block diagram. (10 Marks)

OR

- 8 a. Design a pneumatic circuit for the sequence A+ B+ A- B- and write a ladder program for the same. (10 Marks)
b. Explain with ladder program a Latch circuit and an internal relay. (10 Marks)

Module-5

- 9 a. Differentiate between ball screw and roller screw. (05 Marks)
b. Discuss a traditional and mechatronics design concepts. (05 Marks)
c. Explain the working of hydrostatic bearing with neat sketch. (10 Marks)

OR

- 10 a. Explain any two most commonly used antifriction guideways in CNC machines. (10 Marks)
b. Design a mechatronic system for automatic car park barrier. (10 Marks)

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18ME744

Seventh Semester B.E. Degree Examination, June/July 2023

Mechatronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain with block diagram elements of mechatronics. (07 Marks)
- b. Explain with neat sketch working of Antilock Braking System (ABS). (08 Marks)
- c. With an example, explain components of measurement system. (05 Marks)

OR

- 2 a. Illustrate working of potentiometer and LVDT. (07 Marks)
- b. Explain principle of working of Hall Effect sensor with an example. (08 Marks)
- c. Comparison between the transducer and sensor. (05 Marks)

Module-2

- 3 a. With neat sketch, explain the process of analog to digital conversion. (06 Marks)
- b. Explain the working capacitive filter. (06 Marks)
- c. Explain the importance of supervisory control and data acquisition (SCADA) in current day industries. (08 Marks)

OR

- 4 a. Explain with neat sketch working of solenoids. (06 Marks)
- b. With neat sketch explain working of variable reluctance stepper motor. (06 Marks)
- c. Explain the components and working of DC servo motor. (08 Marks)

Module-3

- 5 a. With block diagram, explain different components and its functionalities of microcontroller. (09 Marks)
- b. List and explain basic elements of control system. (06 Marks)
- c. Differentiate between microprocessor and microcontroller. (05 Marks)

OR

- 6 a. With neat sketch, explain architecture of Intel's 8085 microprocessor. (10 Marks)
- b. Explain the process of Fetch cycle in INTEL 8085 microprocessor. (05 Marks)
- c. Explain the functioning of interrupts in 8085 microprocessor. (05 Marks)

Module-4

- 7 a. With block diagram, explain components and working of basic PLC structure. (08 Marks)
- b. Outline the guidance for selection of PLC for an application. (07 Marks)
- c. Draw and explain ladder diagram for temperature control system. (05 Marks)

OR

- 8 a. Draw ladder diagram and explain extending and retraction of pneumatic piston using latches. (09 Marks)
- b. Explain the control of conveyer motor with ladder diagram. (07 Marks)
- c. Explain with ladder diagram working of internal relay. (04 Marks)

Module-5

- 9 a. Explain construction and working of any one linear motion guideway with neat sketch. (07 Marks)
- b. With neat sketch, explain working of hydro-dynamic bearing used in CNC machines. (05 Marks)
- c. With block diagram, explain Adaptive Control for machine tools with constraints. (08 Marks)

OR

- 10 a. With block diagram, explain elements of closed loop control system. (05 Marks)
- b. How mechatronics design process adopts multi-disciplinary approach in design process? Explain. (08 Marks)
- c. List components of pick and place Robot and explain design consideration made while selecting each component. (07 Marks)

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18ME751

Seventh Semester B.E. Degree Examination, June/July 2023

Energy and Environment

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define energy and power. Explain forms of energy. Definition : Energy : Ability or capacity to do work. (10 Marks)
- b. List and explain factors affecting India's energy development. (10 Marks)

OR

- 2 a. Explain energy production and trade. (10 Marks)
- b. Explain key energy trends in India. (10 Marks)

Module-2

- 3 a. List and explain thermal energy storage methods. (10 Marks)
- b. Explain demographics policy with social and environmental aspects. (10 Marks)

OR

- 4 a. Explain economic analysis of an investment project. (10 Marks)
- b. Explain energy audit and methodology with respect to process Industries. (10 Marks)

Module-3

- 5 a. Define environmental with multidisciplinary nature of environment. (10 Marks)
- b. Define ecosystem with structure and function of ecosystem. (10 Marks)

OR

- 6 a. Explain need for public awareness about environment. (10 Marks)
- b. Explain ecological pyramid with suitable neat sketch. (10 Marks)

Module-4

- 7 a. List and explain causes and control measures of air pollution. (10 Marks)
- b. List the pollution case study. Briefly explain any one case study. (10 Marks)

OR

- 8 a. Explain role of individual in prevention of pollution. (10 Marks)
- b. Explain thermal pollution and nuclear Hazards. (10 Marks)

Module-5

- 9 a. Explain acid rain and causes of acid rain with suitable chemical reactions. (10 Marks)
- b. List and explain the methods of waste land reclamation. (10 Marks)

OR

- 10 a. Explain about air pollution and prevention acts in India. (10 Marks)
- b. Explain issues involved in enforcement of environmental legislation. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg, 42+8 = 50, will be treated as malpractice.

Third Semester B.E. Degree Examination, Jan./Feb. 2023
Transform Calculus, Fourier Series and Numerical
Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Find the Laplace transform of $te^{2t} \frac{2 \sin 3t}{t}$. (06 Marks)

- b. Given that $f(t) = \begin{cases} B, & 0 < t < \frac{\gamma}{2} \\ E, & \frac{\gamma}{2} < t < a \end{cases}$

where $f(t+a) \equiv f(t)$ show that $L\{f(t)\} = \frac{E}{s} \tan h\left(\frac{\pi s}{4}\right)$. (07 Marks)

- c. Using convolution theorem obtain the inverse Laplace transform of the following function : (07 Marks)

$$\frac{(s-1)(s^2+1)}{(s+5)}$$

- 2 a. Find the inverse Laplace transform of: (06 Marks)

$$\frac{s^2-6s+13}{s+5}$$

- b. Express the following function in terms of unit step function and hence find their Laplace transform. (07 Marks)

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ t^2, & 1 < t \leq 2 \\ 0, & t > 2. \end{cases}$$

- c. Solve the following initial value problem by using Laplace transform : (07 Marks)

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = e^{-t}, \quad y(0) = 0, \quad y'(0) = 0$$

Module-2

- 3 a. Obtain Fourier series of $f(x) = \frac{\pi-x}{2}$ in $0 < x < 2\pi$. Hence deduce that (06 Marks)

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$$

- b. Find a cosine Fourier series for $f(x) = (x-1)^2, 0 \leq x \leq 1$. (07 Marks)

- c. Obtain the Fourier series of y upto the first harmonic for the following values. (07 Marks)

x°	45	90	135	180	225	270	315	360
y	4.0	3.8	2.4	2.0	-1.5	0	2.8	3.4

1 of 3

OR

- 4 a. Obtain Fourier series for (06 Marks)

$$f(x) = \begin{cases} \pi x & \text{in } 0 \leq x \leq 1 \\ \pi(2-x) & \text{in } 1 \leq x \leq 2 \end{cases}$$

- b. Obtain the sine half range series for the function: (07 Marks)

$$f(x) = 1 - \left(\frac{x}{\pi}\right) \ln \quad 0 \leq x \leq \pi.$$

- c. The following values of y and x are given. Find Fourier series of upto first harmonics. (07 Marks)

x	0	2	4	6	8	10	12
y	9.0	18.2	24.4	27.8	27.5	22.0	9.0

Module-3

- 5 a. If $f(x) = \begin{cases} 1-x^2, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$. Find Fourier transform of $f(x)$ and hence find the value of (06 Marks)

$$\int_0^\infty x \cos x \cdot e^{-\sin x} dx.$$

- b. Find the Fourier sine transform of $f(x) = e^{-|x|}$ and hence evaluate (07 Marks)

$$\int_0^\infty \frac{x \sin mx}{1+x^2} dx, \quad m > 0.$$

- c. Solve by using Z-Transforms $U_{n+2} + 2U_{n+1} + U_n = n$ with $U_0 = 0 = U_1$. (07 Marks)

OR

- 6 a. Obtain the Fourier cosine transform of the function: (06 Marks)

$$f(x) = \begin{cases} 4x, & 0 < x < 1 \\ 4-x, & 1 < x \leq 4 \\ 0, & x > 4. \end{cases}$$

- b. Obtain the Z-transform of $\cos n\theta$ and $\sin n\theta$. (07 Marks)

- c. Compute the inverse Z-transform of $\frac{3z^2-2z}{(5z-1)(5z+2)}$. (07 Marks)

Module-4

- 7 a. Classify the following partial differential equations: (10 Marks)

i) $x^2 u_{xx} + (1-y^2) u_{yy} = 0, -\infty < x < \infty, -1 < y < 1$

ii) $(1+x^2) u_{xx} + (5+2x^2) u_x + (4+x^2) u_y = 0$

iii) $(x+1) u_{xx} - 2(x+2) u_{xy} + (x+3) u_{yy} = 0$

- b. Solve $u_t = u_{xx}$ subject to the conditions $u(0, t) = 0 = u(1, t)$ and $u(x, 0) = \sin(\pi x)$ by taking $h = 0.2$ for 5 levels. Further write down the following values from the table (10 Marks)

- i) $u(0.2, 0.04)$
ii) $u(0.4, 0.08)$
iii) $u(0.6m, 0.06)$

(10 Marks)

2 of 3

OR

8. a. Solve the elliptic equation $u_{xx} + u_{yy} = 0$ for the following square Mesh with boundary values as shown. Find the iterative values of u_4 (to 9) to the nearest integer.



Fig. Q8(a)

- b. Solve $25u_x = u_t$ at the pivotal points given $u(0, t) = 0 = u(5, t)$, $u(x, 0) = 0$ and $u(x, 0) = \begin{cases} 20x, & 0 \leq x \leq 1 \\ 5(5-x), & 1 \leq x \leq 5 \end{cases}$ by taking $h = 1$ compute $u(x, t)$ for $0 \leq t \leq 1$. (10 Marks)

Module-5

9. a. Given $y'' - xy' - y = 0$ with the initial conditions $y(0) = 1, y'(0) = 0$ compute $y(0.2)$ using fourth order Runge-Kutta method. (06 Marks)
 b. Derive the Euler's equation. (07 Marks)
 c. Find the extremal of the functional $\int_{x_1}^{x_2} (y'^2 + y^2 + 2ye^x) dx$. (07 Marks)

OR

10. a. Obtain the solution of the equation $2 \frac{d^2 y}{dx^2} = 4x + \frac{dy}{dx}$ by computing the value of $y(1.4)$ by applying Milne's method using following data :

x	1	1.1	1.2	1.3
y	2	2.2156	2.4649	2.7514
y'	2	2.3178	2.6725	3.0657

(06 Marks)

- b. Find the curve for which the functional $\int_0^1 [y'^2 + 12xy] dx$ with $y(0) = 0$ and $y(1) = 1$ can be determined. (07 Marks)
 c. Prove that the shortest distance between two points in a plane is straight line. (07 Marks)

CBCS SCHEME

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21ME32

Third Semester B.E. Degree Examination, Jan./Feb. 2023 Metal Casting, Forming And Joining Processes

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define pattern and explain with neat sketches any four pattern allowances. (10 Marks)
b. Explain with a neat sketch sand slinger. (10 Marks)

OR

- 2 a. Explain with a neat sketch investment moulding process. (10 Marks)
b. Explain in detail the procedure to determine the permeability member of green sand in foundry lab. (10 Marks)

Module-2

- 3 a. Explain with a neat sketch cupola furnace showing different zones. (10 Marks)
b. Explain with a neat sketch coreless induction furnace. (10 Marks)

OR

- 4 a. Explain with a neat sketch continuous casting process. (10 Marks)
b. Explain with a neat sketch any five casting defects. (10 Marks)

Module-3

- 5 a. Explain the following yield criteria :
i) Tresca yield criterion ii) Von Mises Yield criterion. (10 Marks)
b. Explain temperature factor in metal forming and also write the comparison between hot working and cold working process. (10 Marks)

OR

- 6 a. Derive an expression for forging pressure and load by slab analysis. (10 Marks)
b. Explain the following sheet metal forming processes with neat sketch.
i) Blanking ii) Piercing iii) Bending. (10 Marks)

Module-4

- 7 a. Explain with a neat sketch Oxy-Acetylene gas welding process. (10 Marks)
b. Explain with a neat sketch types of flames produced in Oxy – Acetylene welding process. (10 Marks)

OR

- 8 a. Explain with a neat sketch Manual metal arc welding and also mention advantages disadvantages and applications. (10 Marks)
b. Explain with a neat sketch Metal Inert Gas (MIG) welding, mention its advantages, disadvantages and applications. (10 Marks)

Module-5

- 9 a. Explain with neat sketch shrinkage in welded structures. (10 Marks)
b. Explain with a neat sketch any five welding defects. (10 Marks)

OR

- 10 a. Write short note for the following : i) Soldering ii) Brazing. (10 Marks)
b. Explain with a neat sketch resistance spot welding process. (10 Marks)

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CBCS SCHEME

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21ME33

Third Semester B.E. Degree Examination, Jan./Feb. 2023 Material Science and Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Classify engineering materials and compare crystalline solids and non-crystalline solids. (07 Marks)
- b. What are voids? Explain Tetrahedral voids and octahedral voids with suitable diagrams. (08 Marks)
- c. Briefly explain the steps to prepare a specimen for microstructural examinations. (05 Marks)

OR

- 2 a. What is meant by imperfections in solids? Explain edge dislocation and screw dislocation. (10 Marks)
- b. In crystallography, what is the theme of symmetry operation? Explain two fold and three fold rotation. (06 Marks)
- c. Define Planar Atomic Density and Atomic Packing Factor. (04 Marks)

Module-2

- 3 a. Define solid solution. Describe the classification of solid solution. (07 Marks)
- b. Draw iron-carbon equilibrium diagram. Explain various phases and locate invariant points. (08 Marks)
- c. Explain the method of measuring hardness in Rockwell Hardness test machine. (05 Marks)

OR

- 4 a. With a neat sketch, explain Fick's 1st and 2nd law of diffusion. (08 Marks)
- b. The solidus and liquidus temperatures for an alloy system containing two metals A and B which are completely soluble in liquid and solid states are presented in the table below. Metal 'A' melts at 1080°C and metal 'B' melts at 1450°C.
 - (i) Construct the phase diagram for the system and label all regions.
 - (ii) Predict the number, type, relative amounts and composition of phases present in an alloy containing 60% A and 40% B at 1250°C.

Sl. No.	Alloy composition (Wt %)	Solidus temp °C	Liquidus temp °C
1	90% A	1100°C	1175 °C
2	60% A	1160°C	1290 °C
3	20% A	1310°C	1400 °C

(12 Marks)

Module-3

- 5 a. Deduce the expression for critical radius of nucleation. (07 Marks)
- b. Draw and explain the process of flame hardening. (06 Marks)
- c. Explain the test procedure of creep test. (07 Marks)

OR

- 6 a. Explain normalizing heat treatment process. Also compare normalizing heat treatment with annealing. (08 Marks)
- b. Explain the concept of mechanisms behind strengthening in metals. Also explain strengthening by grain size reduction. (07 Marks)
- c. Explain Hardening heat treatment in brief. (05 Marks)

Module-4

- 7 a. Explain common types of coatings techniques in brief. (10 Marks)
- b. Briefly explain different types of mechanical methods of powder production techniques. (10 Marks)

OR

- 8 a. What is meant by sintering process? Explain liquid phase and activated sintering. (07 Marks)
- b. How the particle size and particle shape influences the characterization of metal powders? (06 Marks)
- c. Explain the wear test and state the important of conducting wear test. (07 Marks)

Module-5

- 9 a. Explain the need for material selection process in design. (05 Marks)
- b. What are the different types of design? Explain each type. (07 Marks)
- c. Explain the procedure for measuring hardness by Brinell's Hardness testing equipment. (08 Marks)

OR

- 10 a. What are the different functional properties of material? Explain thermal properties. (06 Marks)
- b. Describe in brief about the factors affecting the selection of materials. (06 Marks)
- c. The following data is noted in a tensile test. Diameter of specimen = 200 mm, extension under a load of 10 kN = 0.035 mm, load at yield point = 110 kN, maximum load = 190 kN. Length of specimen after failure = 255 mm, Neck dia = 12.25 mm. Determine :
- (i) Young's modulus (ii) Yield stress (iii) Ultimate stress
- (iv) Percentage elongation (v) Percentage reduction in area (08 Marks)

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CBCS SCHEME

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21ME34

Third Semester B.E. Degree Examination, Jan./Feb. 2023 Thermodynamics

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use of thermodynamic data hand book is permitted.

Module-1

- 1
 - a. State zeroth law of thermodynamics. Explain its significance. (06 Marks)
 - b. What are the similarities and dissimilarities between work transfer and heat transfer? (06 Marks)
 - c. A cylinder contains 1 kg of a certain fluid at an initial pressure of 20 bar. The fluid is allowed to expand reversibly behind a piston according to a law $PV^2 = \text{constant}$ until the volume is doubled. The fluid is then cooled reversibly at constant pressure until the piston regains its original position heat is then supplied reversibly with the piston firmly locked in position until the pressure rises to the original value of 20 bar. Calculate the net work done by the fluid for an initial volume of 0.05 m^3 and draw a neat PV diagram. (08 Marks)

OR

- 2
 - a. Define Heat and Work from thermodynamic point of view. (06 Marks)
 - b. A temperature scale of certain thermometer is given by the relation $t = a \ln p + b$ where a and b are constants and p is the thermometric property of the fluid in the thermometer. If at the ice point and steam point the thermometric properties are found to be 1.5 and 7.5 respectively. What will be the temperature corresponding to the thermometric property of 3.5 on Celsius scale. (06 Marks)
 - c. Apply steady flow energy equation to each of the following: (08 Marks)
 - (i) Nozzle
 - (ii) Boiler
 - (iii) Turbine
 - (iv) Pump

Module-2

- 3
 - a. Prove that Kelvin-Planck statement and Clausius statements of second law of thermodynamic are equivalent. (10 Marks)
 - b. Two reversible heat engines A and B are arranged in series, A rejecting heat to B through the intermediate reservoir. Engine A receives 2000 kJ at a temperature of 421°C from a heat source, while engine B is in communication with a cold sink at a temperature of 4.4°C . If work output of A is twice that of B. Find : (10 Marks)
 - (i) The intermediate temperature between A and B
 - (ii) Efficiency of each engine
 - (iii) The heat rejected to the cold sink

OR

- 4
 - a. State and explain the Carnot cycle with PV and TS diagram. (06 Marks)
 - b. Obtain a relation between COP's of a refrigerator and heat pump. (04 Marks)
 - c. Define entropy. State and prove Clausius inequality. (10 Marks)

Module-3

- 5 a. Explain the following:
 b. (i) Generalized compressibility chart
 (ii) Law of corresponding states
 (iii) Compressibility factor (10 Marks)
 One kg of CO₂ has a volume of 1 m³ at 100°C. compute the pressure by:
- 6 a. (i) Vander Waal's equation (ii) Perfect gas equation (10 Marks)

OR

- b. Explain the following terms with reference to a combustion process:
 (i) Enthalpy of formation (ii) Adiabatic flame temperature
 (iii) Enthalpy of combustion (iv) Heat of reaction (08 Marks)
 Methane is burned with atmospheric air. The analysis of the products on a dry basis is as follows: CO₂ = 10%, O₂ = 2.37%, CO = 0.53%, N₂ = 87.10%
- c. (i) Determine the combustion equation
 (ii) Calculate the air-fuel ratio
 (iii) Percentage theoretical air (08 Marks)
- 7 a. Write Maxwell relations and explain the terms involved. (04 Marks)
 b.

Module-4

- c. Explain P-T diagram for water. (06 Marks)
 Explain the working of a practical regenerative Rankine cycle and derive the efficiency of the cycle. (08 Marks)
 In a steam power cycle, the steam supply is at 15 bar and dry and saturated. The condenser pressure is 0.4 bar. Calculate the Carnot and Rankine efficiencies of the cycle. Neglect pump work. (06 Marks)
- 8 a.

OR

- b. Explain reheat vapour cycle and derive an expression for the reheat cycle efficiency, state the advantages. (10 Marks)
 A vessel having a capacity of 0.05 m³ contains a mixture of saturated water and saturated steam at a temperature of 245°C. The mass of the liquid present is 10 kg. Find the following:
 (i) The pressure (ii) The mass (iii) Specific volume
 9 a. (iv) Specific enthalpy (v) Specific entropy (vi) Specific internal energy (10 Marks)

Module-5

- b. Derive the expression for the air standard efficiency of a diesel cycle. State the assumptions made. (10 Marks)
- 10 a. Explain the different methods of improving the efficiency of Brayton cycle. (10 Marks)

OR

- Air enters the compressor of a gas turbine plant operating on Brayton cycle at 101.325 kPa, 27°C. The pressure ratio in the cycle is 6. Calculate the max temperature in the cycle and the cycle efficiency. Assume $W_T = 2.5 W_C$, where W_T = turbine work, W_C = compressor work. Take $\gamma = 1.4$. (10 Marks)
 Derive the expression for M.E.P of Otto cycle. (10 Marks)

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Question Paper Version : A

Third Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Fill in the blank by translating the given English word to Kannada :

1. You : _____
a) neenu b) Naanu c) yaaru d) Ivaru
2. His : _____
a) NeeVu b) avaru c) aVaLu d) avana
3. What : _____
a) idu b) adu c) yenu d) yelli
4. Friend : _____
a) Gumpu b) Maava c) Sneheta d) GurugaLu
5. Drama : _____
a) naataka b) Kate c) haaDu d) Yaavadu alla
6. Choose the Mathematical + Symbol in Kannada
a) Kale b) Koodu c) Jodisu d) Bhagisu
7. Country in Kannada will be called as
a) D'wesha b) Vesha c) Desha d) Vishesha

Note : Substitute the word from the following each sentence in appropriate places :

8. idu : idakke then adu _____
a) adakke b) idakke c) Horakke d) Horakkee
9. UUru : Urige then avaru _____
a) ivarige b) avarige c) Nimage d) Nawage

10. Mane : maneyalli then Pustaka
a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
11. Raja : Rajanaddu, then Raani _____
a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Write the English word for the given Kannada word :

12. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
13. pramaaNu : _____
a) promise b) very good c) purpse d) Thankyou
14. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
15. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
16. biLi : _____
a) red b) Black c) White d) Green
17. doDDa : _____
a) Big b) Small c) fat d) thin
18. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
19. gurugaLu
a) Former b) Teacher c) Father d) Uncle
20. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast
21. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give
22. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
23. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
24. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
25. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
26. huTTu : _____
a) Good b) die c) Birth d) bed

27. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
28. daKShiNa : _____
a) West b) East c) South d) North
29. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
30. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
31. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
32. granthalaya : _____
a) House b) office c) college d) Library
33. baNNa : _____
a) White b) Banana c) Colour d) Red

Note : Translate the English question to Kannada :

34. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
35. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
36. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla
37. When are you going to College?
a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
38. Had your breakfast?
a) YaaVagu Tindi? b) TinDi ayitaa?
c) oota ayta? d) TinDi yestu beku?

Note : Translate the Question Kannada to English :

39. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?
40. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none

41. ethaKKe hogidde?
 a) Why did you go? b) Where did you go?
 c) When you are going? d) None of these
42. nimma Kalejinaa hesarenu?
 a) What is your college name? b) Where is your college
 c) How old is your college? d) Which college are you studying?
43. naanu yenu MaDabhahudu?
 a) What did you do? b) What should we do?
 c) What can I do? d) None
44. eegale baa
 a) come immediatly b) Go immediatly c) come Tomorrow d) None
45. dayamaDi niiVu oLagaDe banni
 a) When you come b) Come inside
 c) Don't come hear d) you please come inside
46. Kallegige raje KoTTi ddare
 a) Holiday is given to college b) Don't come to college
 c) Today college is there d) None
47. Raamanu nanaginta doDDavanu
 a) Rama is older than me b) Rama is elder than me
 c) Rama is my friend d) None
48. VandneguLu Saar
 a) Very good b) Thank you sir
 c) May I come Sir d) Good morning Sir
49. baalehaNNu
 a) Mango b) Chicku c) Banana d) Orange
50. ToTa _____
 a) Room b) Bangle c) House d) Garden

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Question Paper Version : B

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
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4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate the English question to Kannada :

1. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
2. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
3. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla
4. When are you going to College?
a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
5. Had your breakfast?
a) YaaVagu Tindi? b) TinDi ayitaa?
c) oota ayta? d) TinDi yestu beku?

Note : Write the English word for the given Kannada word :

6. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
7. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
8. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep

9. huTTu : _____
a) Good b) die c) Birth d) bed
10. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
11. daKShiNa : _____
a) West b) East c) South d) North
12. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
13. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
14. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
15. granthalaya : _____
a) House b) office c) college d) Library
16. baNNa : _____
a) White b) Banana c) Colour d) Red
17. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
18. pramaaNa : _____
a) promise b) very good c) purpse d) Thankyou
19. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
20. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
21. biLi : _____
a) red b) Black c) White d) Green
22. doDDa : _____
a) Big b) Small c) fat d) thin
23. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
24. gurugaLu
a) Former b) Teacher c) Father d) Uncle
25. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast

26. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give

27. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair

Note : Substitute the word from the following each sentence in appropriate places :

28. idu : idakke then adu _____
a) adakke b) idakke c) Horakke d) Horakkee

29. UURu : Urige then avaru _____
a) ivarige b) avarige c) Nimage d) Nawage

30. Mane : maneyalli then Pustaka
a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli

31. Raja : Rajanaddu, then Raani _____
a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Translate the Question Kannada to English :

32. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?

33. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none

34. ethaKKe hogidde?
a) Why did you go? b) Where did you go?
c) When you are going? d) None of these

35. nimma Kalejinaa hesarenu?
a) What is your college name? b) Where is your college
c) How old is your college? d) Which college are you studying?

36. naanu yenu MaDabhahudu?
a) What did you do? b) What should we do?
c) What can I do? d) None

37. eegale baa
a) come immediatly b) Go immediatly c) come Tomorrow d) None

38. dayamaDi niiVu oLagaDe banni
a) When you come b) Come inside
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39. Kallegige raje KoTTi ddare
a) Holiday is given to college b) Don't come to college
c) Today college is there d) None

40. Raamanu nanaginta doDDavanu
 a) Rama is older than me b) Rama is elder than me
 c) Rama is my friend d) None
41. VandneguLu Saar
 a) Very good b) Thank you sir
 c) May I come Sir d) Good morning Sir
42. baalehaNNU
 a) Mango b) Chicku c) Banana d) Orange
43. ToTa _____
 a) Room b) Bangle c) House d) Garden

Fill in the blank by translating the given English word to Kannada :

44. You : _____
 a) neenu b) Naanu c) yaaru d) Ivaru
45. His : _____
 a) NeeVu b) avaru c) aVaLu d) avana
46. What : _____
 a) idu b) adu c) yenu d) yelli
47. Friend : _____
 a) Gumpu b) Maava c) Sneheta d) GurugaLu
48. Drama : _____
 a) naataka b) Kate c) haaDu d) Yaavadu alla
49. Choose the Mathematical + Symbol in Kannada
 a) Kale b) Koodu c) Jodisu d) Bhagisu
50. Country in Kannada will be called as
 a) D'weshu b) Vesha c) Desha d) Vishesha

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Question Paper Version : C

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
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Note : Write the English word for the given Kannada word :

1. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
2. pramaaNu : _____
a) promise b) very good c) purpse d) Thankyou
3. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
4. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
5. biLi : _____
a) red b) Black c) White d) Green
6. doDDa : _____
a) Big b) Small c) fat d) thin
7. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
8. gurugaLu
a) Former b) Teacher c) Father d) Uncle
9. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast

10. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give
11. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
12. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
13. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
14. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
15. huTTu : _____
a) Good b) die c) Birth d) bed
16. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
17. daKShiNa : _____
a) West b) East c) South d) North
18. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
19. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
20. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
21. granthalaya : _____
a) House b) office c) college d) Library
22. baNNa : _____
a) White b) Banana c) Colour d) Red

Note : Translate the Question Kannada to English :

23. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?
24. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none
25. ethaKKe hogidde?
a) Why did you go? b) Where did you go?
c) When you are going? d) None of these

26. nimma Kalejinaa hesarenu?
 a) What is your college name? b) Where is your college
 c) How old is your college? d) Which college are you studying?
27. naanu yenu MaDabhahudu?
 a) What did you do? b) What should we do?
 c) What can I do? d) None
28. eegale baa
 a) come immediatly b) Go immediatly c) come Tomorrow d) None
29. dayamaDi niiVu oLagaDe banni
 a) When you come b) Come inside
 c) Don't come hear d) you please come inside
30. Kallegige raje KoTTi ddare
 a) Holiday is given to college b) Don't come to college
 c) Today college is there d) None
31. Raamanu nanaginta doDDavanu
 a) Rama is older than me b) Rama is elder than me
 c) Rama is my friend d) None
32. VandneguLu Saar
 a) Very good b) Thank you sir
 c) May I come Sir d) Good morning Sir
33. baalehaNNu
 a) Mango b) Chicku c) Banana d) Orange
34. ToTa _____
 a) Room b) Bangle c) House d) Garden
- Fill in the blank by translating the given English word to Kannada :**
35. You : _____
 a) neenu b) Naanu c) yaaru d) Ivaru
36. His : _____
 a) NeeVu b) avaru c) aVaLu d) avana
37. What : _____
 a) idu b) adu c) yenu d) yelli
38. Friend : _____
 a) Gumpu b) Maava c) Sneheta d) GurugaLu
39. Drama : _____
 a) naataka b) Kate c) haaDu d) Yaavadu alla

40. Choose the Mathematical + Symbol in Kannada
 a) Kale b) Koodu c) Jodisu d) Bhagisu
41. Country in Kannada will be called as
 a) D'weshu b) Vesha c) Desha d) Vishesha

Note : Translate the English question to Kannada :

42. Where is he?
 a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
43. Where shall we meet?
 a) NaVu yaaru? b) Sanje barattiya?
 c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
44. When is the physics class?
 a) physics class yaavagaa? b) Sanje physicedeyan?
 c) physic Lab elli ede? d) yaavadu alla
45. When are you going to College?
 a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
 c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
46. Had your breakfast?
 a) YaaVagu Tindi? b) TinDi ayitaa?
 c) oota ayta? d) TinDi yestu beku?

Note : Substitute the word from the following each sentence in appropriate places :

47. idu : idakke then adu _____
 a) adakke b) idakke c) Horakke d) Horakkee
48. UUru : Urige then avaru _____
 a) ivarige b) avarige c) Nimage d) Nawage
49. Mane : maneyalli then Pustaka
 a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
50. Raja : Rajanaddu, then Raani _____
 a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

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[illegible]

Question Paper Version : D

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate the Question Kannada to English :

1. eshTu ganTe Saar?
a) Can I go Sir?
b) will I come Sir?
c) 8 Clock Sir
d) What is the time Sir?
2. Ee pustaKa yaaraduu?
a) Is that book yours
b) whose book is this?
c) This book belongs to me.
d) none
3. ethaKKe hogidde?
a) Why did you go?
b) Where did you go?
c) When you are going?
d) None of these
4. nimma Kalejinaa hesarenu?
a) What is your college name?
b) Where is your college
c) How old is your college?
d) Which college are you studying?
5. naanu yenu MaDabhahudu?
a) What did you do?
b) What should we do?
c) What can I do?
d) None
6. eegale baa
a) come immediatly
b) Go immediatly
c) come Tomorrow
d) None
7. dayamaDi niiVu oLagaDe banni
a) When you come
b) Come inside
c) Don't come hear
d) you please come inside

8. Kallegige raje KoTTi ddare
a) Holiday is given to college b) Don't come to college
c) Today college is there d) None
9. Raamanu nanaginta doDDavanu
a) Rama is older than me b) Rama is elder than me
c) Rama is my friend d) None
10. VandneguLu Saar
a) Very good b) Thank you sir
c) May I come Sir d) Good morning Sir
11. baalehaNNu
a) Mango b) Chicku c) Banana d) Orange
12. ToTa _____
a) Room b) Bangle c) House d) Garden

Fill in the blank by translating the given English word to Kannada :

13. You : _____
a) neenu b) Naanu c) yaaru d) Ivaru
14. His : _____
a) NeeVu b) avaru c) aVaLu d) avana
15. What : _____
a) idu b) adu c) yenu d) yelli
16. Friend : _____
a) Gumpu b) Maava c) Sneheta d) GurugaLu
17. Drama : _____
a) naataka b) Kate c) haaDu d) Yaavadu alla
18. Choose the Mathematical + Symbol in Kannada
a) Kale b) Koodu c) Jodisu d) Bhagisu
19. Country in Kannada will be called as
a) D'wsha b) Vesha c) Desha d) Vishesha

Note : Translate the English question to Kannada :

20. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
21. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
22. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla

23. When are you going to College?
 a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
 c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
24. Had your breakfast?
 a) YaaVagu Tindi? b) TinDi ayitaa?
 c) oota ayta? d) TinDi yestu beku?

Note : Substitute the word from the following each sentence in appropriate places :

25. idu : idakke then adu _____
 a) adakke b) idakke c) Horakke d) Horakkee
26. UURu : Urige then avaru _____
 a) ivarige b) avarige c) Nimage d) Nawage
27. Mane : maneyalli then Pustaka
 a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
28. Raja : Rajanaddu, then Raani _____
 a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Write the English word for the given Kannada word :

29. HaNNU : _____
 a) seeds b) flower c) Tree d) Fruit
30. pramaaNu : _____
 a) promise b) very good c) purpse d) Thankyou
31. tangi : _____
 a) Aunti b) Uncle c) Sister d) Mother
32. gaNita : _____
 a) Biology b) Physics c) English d) Mathematics
33. biLi : _____
 a) red b) Black c) White d) Green
34. doDDa : _____
 a) Big b) Small c) fat d) thin
35. Aarogya : _____
 a) Tomarrow b) Health c) wealth d) cold
36. gurugaLu
 a) Former b) Teacher c) Father d) Uncle
37. uuTa : _____
 a) Meals b) Drinks c) Get up d) breakfast
38. KoDabeDa : _____
 a) Give me b) Give us c) give d) Don't give

39. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
40. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
41. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
42. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
43. huTTu : _____
a) Good b) die c) Birth d) bed
44. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
45. daKShiNa : _____
a) West b) East c) South d) North
46. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
47. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
48. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
49. granthalaya : _____
a) House b) office c) college d) Library
50. baNNa : _____
a) White b) Banana c) Colour d) Red

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Question Paper Version : A

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023
Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners on the OMR sheets are strictly prohibited.**

ಸಂಕೇತಿಸಿದ ಉತ್ತರಗಳನ್ನು ಹೀಗೆ ಉತ್ತರಿಸಿ :

1. 9983f16 ಲಿಂಗದೊಳಗೆ 2x ಸ23. 8S3ಯ ಲಿಂಗದೊಳಗೆ ಏನು ರೇಖೆಯು?
a) ಕೃಷ್ಣಾಕ್ಷರಿಸಾರ್ಥ b) ಬುಕ್ಕಾರ್ಥ c) ಹಕ್ಕಬುಕ್ಕ d) ಜಿರಿಸಾರ್ಥ.
2. e6ಸ9n' ಫಿಂಡು 2ಫಿಂಡು----- ಲಿಂಗ, ಒಬ್ಬ, ಸಂಖ್ಯೆ, 9ಗುಂಪಿಯು.
a) ಸಂಖ್ಯೆ b) ಲಿಂಗ c) ಪದ, ದೀರ್ಘ d) ಬಹುಪದ.
3. ಕವಿಗಳಿಂದ ಏನು, ಏನು'ಯಲ್ಲಿ ಏನು'ಯಲ್ಲಿ ಏನು'ಯಲ್ಲಿ?
a) ಏನು'ಯಲ್ಲಿ b) ಕೆಳಗಡೆ c) ಒಂಟಿ d) 2ಅಂಕ.
4. ಸಮೀಪದ, ಸಮೀಪದ ಬಗ್ಗೆ ಕೆಳಗೆ.
a) ದೊಂಟಿ b) ದಸಾ c) ಕುಂದು ಲಿಂಗ d) 2'ನು ಏನು' Are'.
5. ದೊಂಟಿ ಕೆಳಗೆ 'ದೊಡ್ಡ ಬಗ್ಗೆ ಪದ, ಏನು?
a) ಲಿಂಗ ಅಂಕ b) ಏನು, ಏನು' ಅಂಕ
c) 2ಲಿಂಗ, 9ಕೆ ಅಂಕ d) ಕೆ ಏನು'.

6. 1800 680 ಬಹುಶಃ ಗ್ರೇಡ್ ಕೇಕ್ಸ್ ಆದಿವಿ.
- a) 8. 1 b) 99ರಲಿಠ ಪಂ8ತ
- c) ಸಃ ಂ27 2vcr d) ಕBrಷ3ಃ ಆರ್ Ûr87.
7. ಪೆS&d ಲೇ'n'-----aY,S6ಂ ರೆಂ6ಂ?
- a) □'rಸ& b) 91 c) ಹಗಂ d) □'ಂಪ.
8. ಸುಫ'ಫæ &dಂvಂ&8rd3 2ಂoj' ಡಂ, 3d 2Aಫodಂ ಅಡುಂ&.
- a) ಬಹ'ನಂ6& b) ಕುouಂ6& c) ಚ2ಂಲಿ6& d) &dಂvಂ&.
9. ಕೃfi6 □ಂS, ಕೃಗಿ ಕತ್ಯ ರೆಂ6ಂ?
- a) ಆಂಡರೆ, b) ಪಂಪ c) ಬಸSaÂ d) ಹಂಪಂ.
10. d& ಪdd eಫx.
- a) ಚod, b) 69 c) ಸೆcಠx d) 3ಠತ, .
11. 3fiಂS□ ಸಹ9d ಫ2x : 3AಸುS□ದಂ
- a) ಸ. ಫ2x b) 23ಂಪ, ಫ2x c) ಪ6ಫ2x d) ಕುಲಫ2x.
12. □ಂorಂa fiಂ888 ರೆಂS ಡೀ2ಂಡುS□ದಂ?
- a) ತ33 b) fiaa c) ಝಾಝಾ d) ಫಾಫಾ.
13. 9ಪDS ಪdd eಫx®'r3ಂ?
- a) □ಂ,ಂH b) ಸ. fix c) ಆY d) 2ಂತು.
14. 9'-rd. 6ಠ, 3S6 ತS6c6ಂ ರೆಂS□ದಂ?
- a) 2ಂಡ, b) 2ಂಸೆc6ಂ c) ಭಂ, S# d) 2ಂಸಿ'Àr3ಹಿ.
15. ಕ3ಕ್ಂಸ6 2cಲ 2'ಸ6ಂ-----.
- a) ಭ62aÂ ವಂರೆಕ b) ಕ3ಕಪ, c) H2ಲಿಪ,ವಂರೆಕ d) 9r6ಪ.

16. ಕೆಳಗಿನಲ್ಲಿರುವ ಒಂದು ಸಂಖ್ಯೆ ----- ಹೊಂದಿಕೆಯಾಗುತ್ತದೆ
- a) ಕೆಳಗೆ b) ಹೊಂದಿಕೆಯಾಗುತ್ತದೆ c) ಸಂಖ್ಯೆ d) ಹೊಂದಿಕೆಯಾಗುತ್ತದೆ
17. ಒಂದು ಸಂಖ್ಯೆ 253 ಸಂಖ್ಯೆ
- a) 253 b) 253 c) 253 d) 253.
18. ಒಂದು ಸಂಖ್ಯೆ 253 ಆದರೆ, 2 ಸಂಖ್ಯೆ, 2 ಸಂಖ್ಯೆ, 2 ಸಂಖ್ಯೆ
- a) 253 b) ಕೆಳಗೆ c) ಹೊಂದಿಕೆಯಾಗುತ್ತದೆ d) ಹೊಂದಿಕೆಯಾಗುತ್ತದೆ.
19. ಒಂದು ಸಂಖ್ಯೆ 96 ತರುವಂತಹ ಒಂದು ಸಂಖ್ಯೆ ----- ದತ್ತಾಂಶದೊಂದಿಗೆ 2 ಸಂಖ್ಯೆ.
- a) 80 b) 90 c) 100 d) 110.
20. 2 ಸಂಖ್ಯೆ Are' ಕೆಳಗೆ ಕೆಳಗೆ.
- a) u'roj', b) 89 c) ಕೆಳಗೆ d) 2 ಸಂಖ್ಯೆ
21. 86 ವರ್ಷ ಕೆಳಗೆ ಸಂಖ್ಯೆ, ಸಂಖ್ಯೆ 253 ಆದರೆ, 2 ಸಂಖ್ಯೆ.
- a) 253 b) 253 c) 253 d) 253.
22. 253 ಪದವಿ ಪದವಿ.
- a) 96 b) 253 c) ತರುವಂತಹ d) 96.
23. ಪದವಿ, 253 ಸಂಖ್ಯೆ 2 ಸಂಖ್ಯೆ?
- a) 66 b) 253 c) 253 d) 253.
24. ಕೆಳಗಿನಲ್ಲಿರುವ ಒಂದು ಸಂಖ್ಯೆ 253 ಸಂಖ್ಯೆ.
- a) a. i. 253 b) 253 c) 253 d) 253.
25. 253 ಸಂಖ್ಯೆ ಸಂಖ್ಯೆ 253 ಸಂಖ್ಯೆ 253 ಸಂಖ್ಯೆ 253 ಸಂಖ್ಯೆ
- a) 253 b) 253 c) 253 d) 253.

c) ಆ26 ಈ26

d) 2n೦v' Coಬ A&93 ಪSxತೆ. **21KSK37**

26. ಕ&' ವೆdd eಢx

- a) ಁA b) □'cr& c) e& d) ಭಞ

27. ಕೃಡ ಫಿ'ಠೃೃೃ 8ffi&ಫಿ Codೃ ಕ'ದSೃ.

- a) ಕು'ಂಪ□ b) 93ಪೆd6ೃ c) Sಚ3□ೃ66ೃ d) 9v'crಫೃ ಫಿ'.

28. 192068ಢೃಫಿಪ□6d8ಢ 3□'d e8'rd3----- --.

- a) ಫಿ&ತ, e8'rd3 b) □ೃon',ಫ e8'rd3
c) ಁ9&rಢ e8'rd3 d) fi,ಂಢಕತx6 e8'rd3.

29. ಕvೃxYಕ ಁ9, d ಢd83 2'ಸ6ೃ ----- --.

- a) 2ೃಸ6ೃ ಁ9, b) 2ofi&m6ೃ ಁ9, c) u'ofಿ&m6ೃ ಁ9, d) ಠೃS□dೃ ಢ

30. "ಹ'ೃೃಢ' ಉ'c&n' ಢೃೃೃಫಿ&GoYೃ, ತೃ'ಠೃಢ', □'ಃ'ೃೃೃಹ&ಫಿ&GoYೃ, dಢ3□' 2ೃ&G
j'rfಿೃೃಫಿ&GoYೃ," adೃ aS6 Sಚ3ೃಢ'.

- a) ಬಸSaÂ b) jೃೃ'2ಢ, c) eಕ'2ೃೃ'j' r9 d) 2ೃ6ಢ, .

31. "rೃಫid ಫ'crಫidಕ'6dn'rಢd n'cಫ್ಢೃೃೃೃೃಫಿಃ'ೃ ಆಫಿ6ೃಢ 2ೃ9ಸಫಿ" Codೃ ಫrS3d 6'ಕ'e'ಢ
ಆddxS3ೃೃಫಿ ಫೃ&d ಕ9.

- a) 63| b) 93| c) ಪಢ3ಫಿ d) ಪಂಪ.

32. ಆಠೃ ಁ9, fi&8ಢ ಬ&□'ಠೃಫಿೃS ಫಿ' ಆಢತ ಫಿ'ಢೃೃ Hr2ೃx9ಸೃಢA6ೃS ಃೃr'ೃd.

- a) 341v'r ಃೃೃr'ೃd b) 342v'r ಃೃೃr'ೃd
c) 343v'r ಃೃೃr'ೃd d) 344v'r ಃೃೃr'ೃd.

33. ಪ&ಸ6 ಪ6ೃಪ' 2ೃಫಿ ಪ, ತ, ೃ ಫrS3d ಸೃಫಿ& 2cಲಕ 23ೃಪ, 3 ಃೃತ6ಫಿ ಪ□'ಠೃೃS
ಪ&ಪಕ. e'ಢr.

- a) &rಱ b) 9rಱ c) ೃ2x d) ಸೃಸೆÂಱ.

34. ಆೃೃೃಕ ಕೃಡದ ಢY ಢಲ 93ಪೆd ಸೃಫಿ, ಹ fi,ಂಢ.

a) 93ಪದ

b) □ ಂ3ಪದ

c) fi6H0 ²೦ಡು

d) ±೦&ತ, .

35. u'ೞೞ9, ಪದ್ಧ eಧx
a) ಅ' crA b) ಅ' crfi c) ದ೦- 'c, °' d) Mಷ8.
36. ಂಸ ಪದ್ಧ ತದS 6cಪ
a) ಕೞೞ 6cಪ b) 6ಕ'ಸ c) 2' d&□ ' d) ತತ, 2 6cಪSಲ೦.
37. ಪ&rb' 3□'ರೞೞÅj' CೞೞS□dೞ ರೞS □ೞಲ ----.
a) ಭcತ□ೞಲ b) ಭ9ಷe□ೞಲ c) Sತx2ೞ3□ೞಲ d) 2æ'nೞಲ.
38. ರೞS 9vೞofid 9H,rಕ6a e'rbಕ6 2HೞÅ
a) vೞfi b) ee'2ೞ& c) e'cಡS d) n'coಡ. 9vೞofi.
39. ತಲರೞಸ86ೞಕೞಡ, eೞ&G 23'r &rತxv'ರ 6ಚv'□ೞ66ೞ.
a) ಕ3ಕೞೞಸ6ೞ b) ಹ&jೞಸ6ೞ c) ಪ□6od6jೞಸ6ೞ d) jೞಸ6ೞ.
40. 2nೞv' Coಬ A693 ಪSxತ ಪ, ಬೞಫS3ೞC ರೞS ಪ, ೞಸ ಕಧ38od ಅ&□'c&ÑeೞÅj' .
a) A&93 b) 2nೞv' c) A&93 vೞ8n' ಪರೞa d) ಆv'□ೞಡು.
41. ಸೞಕ□' ' 2'd& ಹ&d n'c rೞರ88p-----ತೞೞ9d
a) ಂA b) C&Gå c) □'cr& d) ಭತÅ
42. e'cಅ' fæ-----ದೞ&ತ ಁrfiೞS□j' .
a) 2ೞ8dಅ' b) 2a' ತa' c) ಕುಹda' d) Bod'.
43. ಕೞÅ3 ಕS8ರ ತaÅv' □'cr2ೞe' ----- --.
a) □ೞeೞfi ae'cÅr b) □'c6æ'm&Ae'cÅr c) □ೞೞೞ d) □'ೞ8D ae'cÅr.
44. ಕ3Cಡd ೞY ೞdಲ Sಚ3nೞHx.
a) H, ೞ'rh b) ದೞfi¶æ' c) ಆರÅ& ಲಕ'2å d) eಕ'22ೞj'r9.

45. 2'cಸ 9fiಂ6ಂ -----ಕೆ86ಲು 26±'cಬfiಂ
a) 2'cಸCe' b) 2'cಸ ಹೇಂ c) 2'cಸ□ಂಂಡ d) ಹಜೆ'u'r6ಂ.
46. 9ಂಠ'n' ಒಷು, ಪ,2ಂb □ಂೌdಲ,fi6ಂತೆÅ'.'
a) 10 b) 4 c) 6 d) 8.
47. ಕು6ಂಡು □ಂಾರಂa ಪd,53ಂç ಅರÅ ಕS3 ಸಂಕಲ3 ----- --.
a) fi& b) vಂd8re' c) ಸ9Arತ d) vಂಕುತಂH.
48. 2nಂv' Coಬ A & ರ 8 ಲಿಂಂಸಂಂA6ಂS 9vಂofi
a) n'coಡ b) ಕುa9 c) ee'2ಂ& d) □ಂಡು93.
49. 9ಂ6ತೆS□ ಬ□',ರ 2r83-----ಕೆe'n' 2cಲ v'e'.'
a) 9ತ, b) ಒÅ c) 2ಂd, a d) v'rರಃ.
50. ಕ,2, &ಸುÅ, ತಕxಬdÆಂd ಆe'crಚv', 9ಪÅಪ್ಪ'e' aS-----6S6 ಹುYo, fiಂafi&G.
a) ಸಸಿ. Co. 9-'.rd.6ರ, b) nಂಂ8rಗ್ಗ
c) e&'2'ುಂ&G 36ರಂಹ 2cHx d) ರಂS□ಂಯ ಲಿ

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Question Paper Version : B

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023
Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

ಸಕಲಾಂಗ ಉತ್ಪಾದನೆಯಲ್ಲಿ ಉತ್ಪಾದನೆಯು :

1. ಸುಂಕ' ' 2' d& ಹ&d n' c r h 8 8 ----- ತುಂ9d
a) ಅಂA b) C&G i c) ' cr& d) ಭತ್ತ್
2. e' c a' f æ ----- ದಯ&ತೆ UrfioyS i' .
a) 2ಂ8d a' b) 2a' ತa' c) ಕುಹd a' d) Bod i'.
3. ಕಹ&3 ಕS88 ತaAv' ' cr2e' ----- .
a) ಲಾಂfi ae' c År b) ' c6æ' m&Ae' c År c) ಲಾಂfi ಣಿ d) ' ಲಾಂfi ae' c År.
4. ಕ3Çಡd uY udಲ Sಚ3ನಂ&Hx.
a) H, ' r h b) ದಂfi ಫಾ' c) ಆರ& ಲಕ'2i d) eಕ'22ಂ; 'r9.
5. 2' c ಸ 9fiಂ6ಂ ----- ಕc86ಲು 26±' cಬfiಂ
a) 2' c ಸCe' b) 2' c ಸ ಹಂ c) 2' c ಸ ಲಂ d) ಹಾ'u'r6ಂ.

- [illegible]

26. ಕೆಳಗಿನಲ್ಲಿ ಒಂದನ್ನು 'ಕೆಳಗಿನ' ----- ಹೊಂದಿರುವ ಒಂದನ್ನು 'R'
 a) ಕೆಳಗಿನ b) ಹೊಂದಿರುವ c) ಸಮಾನ d) ಹೊಂದಿರುವ
27. ಒಂದನ್ನು '2' ಸಮಾನ
 a) ead b) edR c) ead e d) aad e.
28. ead, 2 'e' ರ 2 'e', R, ead, ead
 a) aad e b) ಕೆಳಗಿನ c) ಹೊಂದಿರುವ d) ಹೊಂದಿರುವ, 2 'e'.
29. ಒಂದನ್ನು '96' ----- ddx3 'e' 2 'e' R.
 a) 8ofi b) 9ofi2 c) fiad d) &S.
30. 'e' ಸಮಾನ 'Are' ಕೆಳಗಿನ ಕೆಳಗಿನ
 a) u' R, b) 89g c) ಕೆಳಗಿನ d) 2 'e'.
31. 9983fi6 ಒಂದನ್ನು '2' ಸಮಾನ. 83 'e' ಒಂದನ್ನು 'e' ರ 2 'e' ರ?
 a) ಕೆಳಗಿನ 'e' b) ಬಹು 'e' c) ಹೊಂದಿರುವ d) 'e'.
32. ead 'e' ಒಂದನ್ನು 'e' ----- 'e', e, e, a 9g, ead.
 a) e, a b) 18e c) e, d e d) ಬಹು 2 'e'.
33. ಕೆಳಗಿನಲ್ಲಿ 'e', 'e' ರ 2 'e' ರ 'e' ರ?
 a) e' e b) ಕೆಳಗಿನ c) e d) 2 'e'.
34. S, ead, ead ಬಹು 'e' ಕೆಳಗಿನ.
 a) ead b) d, Sa c) ಕೆಳಗಿನ ead d) 'e' ಸಮಾನ 'Are'.
35. ead ಕೆಳಗಿನ 'e' ಬಹು 'e', ead
 a) 19e ead b) e, 'e' ead
 c) 2 'e', e ead d) ಕೆಳಗಿನ ead.

36. 1800 680 ಬಹುಶಃ 'ಗ್ರೇಡ್ ಕೇಕ್ಸ್ ಆಡಾಸಿಂಗು.
- a) 8. ರೆ b) 99ರಲಿರಿ ಪಂ8ತೆ
c) ಸಿ ಂ27 2vcr d) ಕBRಷ35 ಆರ್ UR87.
37. ಪೆS&d ಲೀ'n'-----aY,S6ಂ ರಿಂ6ಂ?
- a) 'ರಸ& b) 91 c) ಹೇಂ d) 'ಂಪ.
38. ಸು4'fæ &ದಂvಂ&8rd3 2ಂoj' ಡಂ, 3d 2A£odಂ ಅಡುಂ&.
- a) ಬಹ'ನಂ6& b) ಕುouಂ6& c) ಚ2ಂ6& d) &ದಂvಂ&.
39. ಕೃfi6 ಂS, ಕೃಹ ಕೃ ರಿಂ6ಂ?
- a) ಆಂಡರೆ, b) ಪಂಪ c) ಬಸSaÂ d) ಹಂಪಂ.
40. d& ಪdd eಢx.
- a) ಚod, b) 69 c) ಸೆಡx d) 3೦ತ, .
41. 86v'R ಕೆ3ಡ ಫಂ&ತ, ಸ2ಬೆ3d eೃ,೦&Ad&S6ಂ.
- a) 2'RT 'cRL b) U, . 'cಡ&6on'Rಂ&ಡಂ
c) 23ಂ ಬಗ್ಗಂ6 d) ರಿಂS&ದಂ ಅ
42. ತೃ' ಪdd eಢx.
- a) 9R6ಂ b) uಂರೆಂ&' c) ತಂಪ d) 91ಲು.
43. ಪ,2ಂ&3 2'ಂಡಹ 2'ಸಾ'ರ3ಂ?
- a) 6cಪ b) 2'R&ಂ c) ಆ6H d) 5ಂ6H.
44. ಕಂ&Yಕ ಫಿತಂ'ಭಸ ಕೃಹ3ಂ& 691dS6ಂ.
- a) a. i. 2'ೈ b) ಅಲೆಂ 2'ಂಕೈ&ರೆಂ
c) u'R0j', d) 2& 2'ಂಕ'RT eರೆ, on&.
45. ಅ%ಕಂ& 2ಂ&ic ಸ2ಂ&ಕಂ& 'ಸ&6d8D6ಂS S,&fi&G ಉಗ್ಗ'coಡು ಬಂ86ಂS
ಫಂಪ,ಂ&ಕ 2ಂ&ಲ,fi& ಹ6ಂ&A6ಂS ಕ'8R
- a) ರಿಂನಂ8 b) ಕಂ&Yಕ ಸಂಸ&H
c) ಆ26 ಈ26 d) 2ನಂv' Coಬ A&93 ಪSxತ.

46. ಕ&' ಪದ eಫx
a) ಅA b) □' CR& c) e&' d) ಭA
47. ಕೆಳಗಡೆ ಟಿಪ್ಪಣಿಯಲ್ಲಿ ಕೊಟ್ಟಿರುವ Codon ಕೆಲವು.
a) ಕು'ಂಪ□ b) 93ವದ6ು c) Sಚ3□ಾ66ು d) 9v' CRಟಾ ಟಿಪ್ಪಣಿ'.
48. 192068Dಂಫಿಪ□6d8D 3□'d e8' Rd3 ----- --.
a) ಟಿಪ್ಪಣಿ, e8' Rd3 b) □ಾon', T e8' Rd3
c) ಅ9&R e8' Rd3 d) fi,ಂಧಕತx6 e8' Rd3.
49. ಕೆಲವು Yಕ ಅಂಶ, d ūd83 2' ಸ6ು ----- --.
a) 2್ಯಸ6ು ಅಂಶ, b) 2ofi&m6ು ಅಂಶ, c) u'ofi&m6ು ಅಂಶ, d) ರಾS□ದು ಆ
50. "ಹೇಂದಾ' ಉಅ' c&n' ಬಂಜರಿ&GoY, ತೈ'ರಂದಾ', □'a'uo೨ಹ&Nfi&GoY, dಠ3□' ಲೂG
j'rfiಯಲಿ&GoY," ಅದು aS6 Sಚ3°ಾAj'.
a) ಬಸSaÂ b) ಜಾಲಿ2, c) ಕೆ'22°aj' R9 d) 2ಾ6ರ, .

USN

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Question Paper Version : C

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023
Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners on the OMR sheets are strictly prohibited.**

ಸಕಲಾಂಶ ಉತ್ತರಿಸುವುದು ಒಂದೇ ಉತ್ತರ :

1. 86v'R ಕೆಳಕಂಡ ಉತ್ತರ, ಸರಿಯಾದದ್ದು ಆಯ್ಕೆಮಾಡಿ.
 - a) 2'RT □ 'cRL
 - b) U, . i'cಡÀ6on'Rnoಡ6ಂ
 - c) 23ಂ ಬರೆಯಿರಿ
 - d) ರೂಪಾಯಿ ೬೫
2. ತ್ಯಂ ಪದದ ಅರ್ಥ.
 - a) 9R6ಂ
 - b) ಉಂಟಾದ್ದು □ '
 - c) ತಂಪು □
 - d) 9೯೮.
3. ಪ, ಒಂದೇ 2' ಒಂದೇ 2' ಸಾ' ರೀತಿ?
 - a) 6cಪ
 - b) ೩' R&ಂ
 - c) ಆ6೫
 - d) ೯ಂ6೫.
4. ಕಂಪ್ಯೂಟರ್ ವಿತರಣೆ ಕೃಷಿಯು 69೯೬6ಂ.
 - a) a. i. ೩' ಂ
 - b) ಆಲೂತಂ *'ಂಕೆಯೊಡೆಂ
 - c) u'R0j',
 - d) 2೩ *'ಂಕೆ □ 'RT e೬,ಂಂ.
5. ಅಕ್ಷರವಾ 2ಂಫಿ ಸಂಪ್ರದೇಶವಾ □ 'ಸÀ6d8D6ಂ S, &fi&G ಉಳಿ'ಂಕೆ ಬಂಕಂ.

ಉಂಟು, ಉಳಿ 2ಂಫಿ, ಫಿ ಹಂಪಾ6ಂ ಕ' ರ

 - a) ರಂಂ
 - b) ಕಂಪ್ಯೂಟರ್ ಸಂಪ್ರದೇಶ

c) ಆ26 ಈ26

d) 2n೦೩v' Coಬ A&93 ಪೆSxತ.

6. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
7. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
8. 1920ರಲ್ಲಿ ಜಾರಿಗೆ ಬಂದ ಕಾನೂನು ಯಾವುದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
9. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
10. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
11. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
12. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
13. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
14. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250
15. ಕೆಳಕಂಡವುಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂಕ 100 b) ಅಂಕ 150 c) ಅಂಕ 200 d) ಅಂಕ 250

- [illegible]

35. ಕೃಷ್ಣಾಸುರ 2ನೇ ಸ್ತಂಭ ----- .
 a) ಭೃಗುನಾಥನು ವಾಚಕ b) ಕೃಷ್ಣಾಸುರ c) ಹೃಷಿಕೇಶನು ವಾಚಕ d) ಭೃಗುನಾಥ.
36. ಕೃಷ್ಣಾಸುರ 2ನೇ ಸ್ತಂಭ ----- ಹೃಷಿಕೇಶನು 2ನೇ ಸ್ತಂಭ 'R
 a) ಕೃಷ್ಣಾಸುರ b) ಹೃಷಿಕೇಶ c) ಹೃಷಿಕೇಶ d) ಹೃಷಿಕೇಶ
37. ಕೃಷ್ಣಾಸುರ 2ನೇ ಸ್ತಂಭ
 a) ಕೃಷ್ಣಾಸುರ b) ಕೃಷ್ಣಾಸುರ c) ಕೃಷ್ಣಾಸುರ d) ಕೃಷ್ಣಾಸುರ.
38. ಕೃಷ್ಣಾಸುರ 2ನೇ ಸ್ತಂಭ ಆದ, 2ನೇ ಸ್ತಂಭ 'c, ಹೃಷಿಕೇಶನು, ಹೃಷಿಕೇಶನು
 a) ಕೃಷ್ಣಾಸುರ b) ಕೃಷ್ಣಾಸುರ c) ಹೃಷಿಕೇಶನು d) ಹೃಷಿಕೇಶನು, ಹೃಷಿಕೇಶನು.
39. ಕೃಷ್ಣಾಸುರ 96ನೇ ಸ್ತಂಭ ----- ddx3ನೇ ಸ್ತಂಭ 2ನೇ ಸ್ತಂಭ 'Rಕು.
 a) 8ನೇ ಸ್ತಂಭ b) 9ನೇ ಸ್ತಂಭ c) 10ನೇ ಸ್ತಂಭ d) &S.
40. 2ನೇ ಸ್ತಂಭ 'Are' ಕೃಷ್ಣಾಸುರ ಕೃಷ್ಣಾಸುರ.
 a) u'Roji', b) 89ನೇ ಸ್ತಂಭ c) ಹೃಷಿಕೇಶನು d) 2ನೇ ಸ್ತಂಭ
41. 99ನೇ ಸ್ತಂಭ ಕೃಷ್ಣಾಸುರ 92ನೇ ಸ್ತಂಭ. ಹೃಷಿಕೇಶನು ಕೃಷ್ಣಾಸುರ ಕೃಷ್ಣಾಸುರ ಕೃಷ್ಣಾಸುರ?
 a) ಕೃಷ್ಣಾಸುರ 'RSಕೃಷ್ಣಾಸುರ b) ಹೃಷಿಕೇಶನು c) ಹೃಷಿಕೇಶನು d) j'RSಕೃಷ್ಣಾಸುರ.
42. ಹೃಷಿಕೇಶನು 2ನೇ ಸ್ತಂಭ ----- 'cಯ, ಒಬ್ಬ, ಹೃಷಿಕೇಶನು, 9ನೇ ಸ್ತಂಭ 'cಯ.
 a) ಹೃಷಿಕೇಶನು b) 18ನೇ ಸ್ತಂಭ c) ಹೃಷಿಕೇಶನು d) ಬಹುಸ್ತಂಭ.
43. ಕೃಷ್ಣಾಸುರ ಕೃಷ್ಣಾಸುರ, ಕೃಷ್ಣಾಸುರ ಕೃಷ್ಣಾಸುರ 'odಯ ಲೇಖಕಿ?
 a) ಕೃಷ್ಣಾಸುರ b) ಕೃಷ್ಣಾಸುರ c) ಹೃಷಿಕೇಶನು d) 2ನೇ ಸ್ತಂಭ.
44. ಸುರೋದ, ಹೃಷಿಕೇಶನು ಬಾ'ದ ಕ ' .
 a) ಹೃಷಿಕೇಶನು b) d.Sa c) ಹೃಷಿಕೇಶನು d) 2ನೇ ಸ್ತಂಭ 'Are'.
45. ಹೃಷಿಕೇಶನು ಕ 'ಹೃಷಿಕೇಶನು ಬಹುಸ್ತಂಭ ಪ್ರಕೃತಿ
 a) 19ನೇ ಸ್ತಂಭ b) 1ನೇ ಸ್ತಂಭ, 1ನೇ ಸ್ತಂಭ 9ನೇ ಸ್ತಂಭ

c) 2□, 9ಕೆ aoಗ್ಗಿಡೆ

d) ಕೆ ಂವೊಡೆ.

21KSK37

46. 1800 680 ಬಾಷೆಗೆ ಹೆಸರು ಕೆಳಕಂಡು. ಆಯ್ಕೆ.
- a) 8. 1' b) 998080 ಪಂಚತ
c) ಸಹಿ 2021 2021 d) ಕೆಳಕಂಡು ಆಯ್ಕೆ.
47. ಪದವಿ ಲೇಖನ -----ಅರ್ಥವು ಹೇಗಿರುತ್ತದೆ?
- a) 1' ಸಹಿ b) 91 c) ಹೇಗಿರುತ್ತದೆ d) 1' ಪಂಚ.
48. ಸಮೀಕ್ಷೆ ಹಾಗೂ 800 2001 ರಿಂದ 2001 ರವರೆಗೆ.
- a) ಬಾಷೆಗೆ ಹೇಗಿರುತ್ತದೆ b) ಕುಳಿತುಕೊಳ್ಳುವುದು c) ಚರ್ಚೆಗೆ ಹೇಗಿರುತ್ತದೆ d) ಹಾಗೂ.
49. ಕೆಳಕಂಡು, ಕೆಳಕಂಡು ಕೆಳಕಂಡು?
- a) ಆಯ್ಕೆ, b) ಪಂಚ c) ಬಸವಾಚಾರ್ಯ d) ಹಂಪಿ.
50. ದೇವರ ಪದವಿ.
- a) ಚರ್ಚೆ, b) 69 c) ಸಹಿ d) 308, .

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Question Paper Version : D

Third Semester B.E./B.Tech Degree Examination, Jan./Feb. 2023
Samskruthika Kannada

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

ಸಕಲಾಂಗ ಉತ್ತರಿಸುವುದು ಒಂದು ಉತ್ತರ :

1. "ರೂಢಿ ಸ್ವರ್ಣದ ಗುಣವು ಒಂದು ನೈಸರ್ಗಿಕ ಲಕ್ಷಣವಾಗಿದೆ" ಎಂದು ಹೇಳಿರುವ ಕವಿ ಯವರ ಹೆಸರು?

a) 63 b) 93 c) ಪುಷ್ಪ d) ಪಂಪ.
2. ಆರಂಭಿಕ ಕಾಲದಲ್ಲಿ ಬಳಸುತ್ತಿದ್ದ ಸಾಧನವು ಯಾವುದು?

a) 341v'r e30r'rd b) 342v'r e30r'rd
c) 343v'r e30r'rd d) 344v'r e30r'rd.
3. ಪುಸ್ತಕವು ಯಾವುದರ ಮೇಲೆ ಪ್ರಭಾವ ಬೀರುತ್ತದೆ?

a) 8R# b) 9R# c) 02x d) ಸಂಸಹ.
4. ಅಧ್ಯಯನ ಕ್ಷೇತ್ರದಲ್ಲಿ ಯಾವುದು ಒಂದು ಪ್ರಮುಖ ವಿಷಯ?

a) 93ಪದ b) 03ಪದ c) 66ಹೆ 20ಡು d) 10ತ.

5. $u'_{\alpha} \neq 0$, $\text{pdd } e_{\alpha}$

a) $\underline{a}'_{\text{CRA}}$ b) $\underline{a}'_{\text{CRfi}}$ c) $d \circ \mathcal{J}^{-1} c, \circ'$ d) $M\overline{\omega}8.$

6. ಅಂಶ ಪದ ತದ್ಗತವು

a) ಕೆಳಗಿನ 6ರ ಪರಿಣಿತರನ್ನು b) 6ರ ಕೆಳಗಿನ c) 2' d & 1' d) ತತ್ 2 6ರ ಪರಿಣಿತರನ್ನು.

7. ಪೆಂಟಾಕ್ಸಾನ್ 'ಶಂಖಾಂಜಿ' C₅H₈O₂ ದು ಶಂಖಾಂಜಿ ಖಾಂಜಿ -----.

a) བཅུ་ལྔ་ལོ། b) བཅུ་དྲུག་ལོ། c) བཅུ་གསུམ་ལོ། d) བཅུ་ལྔ་ལོ།

[illegible]

a) v೦ಫಿ b) ee'2೦ಃ c) e' cಡೆS d) n' coಡ. 9v೦ಫಿ.

9. ತಲದಿಹಿಸುಕಂಡ, ಎಂ&G 23'R &RತXV'ಠ 6ಜv'□ಂ66ಂ.

a) ಕೆಳಗಿನಂತೆ b) ಹಗ್ಗಿನಂತೆ c) ಪೆಟ್ಟಿನಂತೆ d) ಇತರ.

10. 2ನಂವ' Coಬ A693 ಪSxತ ಪ, ಬಂಠS3ಯಿೞಿ ಶಂS ಪ, ಿಂಸ ಕಢ38od ಆೞಿ' c&ÑೆಂAಿ'.

a) $A \propto \frac{1}{v}$ b) $2n \propto v'$
c) $A \propto \frac{1}{v}$ ಎಂದರೆ $\lambda \propto v'$ d) $A \propto \frac{1}{v}$ ಎಂದು.

11. 9983fi6 □_oal d8D σ2x ಸ23. 8S3_ುÇ □_oal d8D a_o9 8_o6_ು?

a) ಕೃಷ್ಣಾಃ ರಸಾಃ b) ಬುಕಾಃ c) ಹಕ'ಬುಕ' d) ಿ'ರಸಾಃ.

12. e6ಸ9n' fioಡು 2fioೃೞೞda'----- □'cYೃ, ಒಬ, ಫೞ, a 9ಙ್ಗೃೞAೞYೃ.

a) ಸ್ತಂಭ, a b) ಲಿಪಿ ' c) ಪು, ದಿ' d) ಬಹುಮಾನ.

13. ಕವಂಖಕೆದ ಅಂಶ, ಟಂಶ' ಡೆಂಚ ಏವ' ಡಂಶ ಂಕ್ರಲಿ'?

a) e'ಲುಫಿಂ b) ಕೆ3ಞಡ c) &೦8 d) 2^೩ಂb.

14. Sಸುಠ' ROD, eS6ು ಬ^a'd ಕೆ '.

a) ರೆಂನಾಂ8 b) dSa c) ಕುಂಯಡು ಲೋರಾ d) 2' cಸೆ ಁಲೆ3 Are'.

15. ಕೆಳಕಂಡ ಕೆಲಸಗಳಲ್ಲಿ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) 19ನೇ ಅಕ್ಟೋಬರ್ b) 19ನೇ, 20ನೇ ಅಕ್ಟೋಬರ್
c) 20ನೇ, 19ನೇ ಅಕ್ಟೋಬರ್ d) ಕೆಳಕಂಡವುಗಳಿಲ್ಲ.
16. 1800 ರಿಂದ 1850 ರ ವರೆಗೆ ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) 8. 1 b) 99ನೇ ಅಕ್ಟೋಬರ್ ಪಂಚಮಿ
c) ಸಹಿ 20ನೇ 2ನೇ ವರ್ಷ d) ಕೆಳಕಂಡವುಗಳಿಲ್ಲ.
17. ಪದವಿ 'ನ' -----ಯ ಸಹಿ ಹೇಗಿರುತ್ತದೆ?
- a) 'ಸಹಿ' b) 91 c) ಹೇಗಿರುತ್ತದೆ d) 'ಸಹಿ'.
18. ಸಹಿ 'ಸಹಿ' & ದಯವಿಟ್ಟು 2ನೇ, 3ನೇ 2ನೇ ದಿನದ ಅಕ್ಷರವನ್ನು ಹೇಗಿರುತ್ತದೆ.
- a) ಬಹುನೇ b) ಕುಂಜನೇ c) ಚಕ್ರವರ್ತಿನೇ d) & ದಯವಿಟ್ಟು.
19. ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಅಂತರ, b) ಪಂಪ c) ಬಹುನೇ d) ಹಂಪನೇ.
20. ದೇವತೆ ಪದವಿ ಹೇಗಿರುತ್ತದೆ.
- a) ಚಕ್ರ, b) 69 c) ಸಹಿ d) 3ನೇ, .
21. ಸಹಿ 'ಸಹಿ' 2ನೇ ದಿನದ ಹೇಗಿರುತ್ತದೆ & 2ನೇ ದಿನದ ಹೇಗಿರುತ್ತದೆ ----- ತುಂಬಿ.
- a) ಅಂತರ b) C&G c) 'ಸಹಿ' d) ಭಕ್ತಿ
22. 'ಸಹಿ' & ----- ದಯವಿಟ್ಟು 2ನೇ ದಿನದ ಹೇಗಿರುತ್ತದೆ.
- a) 2ನೇ ದಿನದ b) 2ನೇ ತಾರೀಖು c) ಕುಂಜನೇ d) Bod'.
23. ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) 'ಸಹಿ' & 'ಸಹಿ' b) 'ಸಹಿ' & 'ಸಹಿ' c) 'ಸಹಿ' & 'ಸಹಿ' d) 'ಸಹಿ' & 'ಸಹಿ'
24. ಕೆಳಕಂಡ ಯಾವುದು ಸರಿಯಾದದ್ದು?
- a) ಸಹಿ, 2ನೇ ದಿನ b) ದಯವಿಟ್ಟು c) ಅಂತರ, 2ನೇ ದಿನ d) 'ಸಹಿ' & 'ಸಹಿ'.

25. 'ಸೆ 9ಫಿಂ6ಂ -----ಕೆ86ಲು 26±'ಸಿಫಿಂ
 a) 'ಸೆಸೆ' b) 'ಸೆ ಹೆಂ b) 'ಸೆಸೆ' c) 'ಸೆಸೆ' d) ಹೆ'ು'6ಂ.
26. 'ನ' ಒಪ್ಪು, ಪ, 2ಂ b ಒಂ ಒಲ, ಫಿಂ6ಂ ತೆ' .
 a) 10 b) 4 c) 6 d) 8.
27. ಕುಂ6ಂ ಒಂ 6ಂ 6ಂ ಪದ, ಸಿಂ6ಂ ಆರೊ ಕೆ3 ಸಂಕಲಿ -----
 a) ಫಿ& b) ವಂ6ಂ8Re' c) ಸೆ9ARತ d) ವಂಕುತಂ.
28. 2ನಂವ' ಕುಂ A & 8 ಉಂ ಸೆ' 6ಂ 6ಂ 9ವಂ6ಂ
 a) ನ'ಕುಂ b) ಕುಂ a9 c) ee'2ಂ& d) ಒಂ6ಂ93.
29. '6ಂ6ಂ ಬ' 6ಂ 2R83 -----ಕೆ'ನ' 2ಕಲಿ v'e' .
 a) 9ತ, b) 6ಂ c) 2ಂd, a d) v'R8.
30. ಕ, 2, & ಸು, ತಕುಂ6ಂ ಆ'ಕೆ'v', 9ಪೆ' aS -----6S6 ಹುಂ, ಫಿಂ6ಂ6ಂ.
 a) ಸೆ. Co. 9'. Rd. 6ಂ, b) ನಂ6ಂ8Rg
 c) e&'ುಂ, & 36ಂಹ 2ಕHx d) ಕೆ'6ಂ6ಂ 6ಂ
31. 86v'R ಕೆ3ಕೆ 6ಂ&ತ, ಸೆ2ಂ&3d eೃ, ಸೆ'6ಂ6ಂ6ಂ.
 a) 2'RJ ಒ'ಕೆ
 b) 6ಂ, . 'ಕೆ'6ಂ6ಂ'6ಂ6ಂ
 c) 23ಂ ಬ'6ಂ6ಂ d) ಕೆ'6ಂ6ಂ 6ಂ
32. ತ್ಯಂ' ಪದ್ಧಿ eೃ.
 a) 9R6ಂ b) ಉಂ6ಂ&ಒ' c) ತಂಪೆ d) 9'ಲು.
33. ಪ, 2ಂ6ಂ6ಂ 2'6ಂ6ಂ 2'ಸೆ'6ಂ6ಂ?
 a) 6ಕೆ b) 6'6ಂ6ಂ c) ಆ6H d) 6ಂ6ಂ.
34. ಕುಂ6ಂ6ಂ ಫಿ'6ಂ6ಂ ಕುಂ6ಂ6ಂ 6ಂ6ಂ6ಂ.
 a) a. i. 6'6ಂ
 b) ಆಲಕುಂ 6'6ಂ6ಂ6ಂ
 c) u'R0j', d) 6ಂ 6'6ಂ6ಂ 6ಂ, 6ಂ6ಂ.

35. ಅ%ಕಠಾ ೨ಫಿ ಸ೨ಠಕಠಾ □' & ಸೆ6d86ಯ S, & ಫಿ & G ಉಳಿ□' ಒಡು ಬಂ86ಯ
ಉಂಪ, ಲಕ ೨ಠಲ, ಫಿ & ಹ6ಯಒಠಾ6ಯ ಕ 'ರ

- a) ರೆಂಟಿ
b) ಕೆಎಚ್‌ಎಸ್ ಸಂಪನ್ಮೂಲ
c) ಆ26 ಈ26
d) 2ನೇ ವರ್ಗದ ಕೆಎಚ್‌ಎಸ್ ಪರಿಷ್ಕರಣೆ.

36. ಕ಼಼' ಪದ್ಧ eಢx

- a) $\frac{1}{2}A$ b) $\frac{1}{2}CR$ c) eR d) $\frac{1}{2}A$

37. ಕೆಳಕಂಡ ಕ್ರಮದಲ್ಲಿ ಕೆಳಕಂಡ ಕೋಡ್‌ಗಳನ್ನು ಕೊಡು.

- a) ಕು^ಠ'ಂಪ□ b) 93ಪದ6ು c) ಸಿಚೆ3□ಾ66ು d) 9v' ರಗ್ತಾ ತಾ^ಠ'.

38. 192068Dvೌಫಿವೃ6d8D 3□'d e8®'Rd3 ----- --.

- a) $\pm 0.03, e8^{\circ} \text{ Rd3}$ b) $\square_{\text{on}}, i^{\circ} e8^{\circ} \text{ Rd3}$
c) $\pm 0.09 \text{ Rd3}, e8^{\circ} \text{ Rd3}$ d) $\text{fi}, \text{on}, \text{d}^{\circ} \text{ Rd3}, e8^{\circ} \text{ Rd3}$.

39. ಕವಂxYಕ ಁಂ9, d ūd83 2' ಸ6ಂ ----- --.

- a) 2000, b) 2001, c) 2002, d) 2003

40. "ಹೌದ್ ಉಂ' ಎನ್' ಬಂತೆಫಿಗೊಯ್, ತ್ಯು' ಹೌದ್', □' ಅ' ಯಾಹ್‌ನಿಗೊಯ್, ದಶೆ' ವಾಗ್
ಇ' ರ್‌ಫಿಯೊಯ್, " ಆಯ್ ಆಸೆ ಸಚೆವಾಗಿ'.

- a) ಬಸSaÂ b) i_ಾl'2ð, c) eಕ'2²_ಾj'_R9 d) 2_ಾ6ð,.

41. $3f \rightarrow 3d$ ಸಹಜ $2x : 3A \rightarrow 3d$

- a) ಸ. ೪2x b) 23ಂಪ, ೪2x c) ಪ6೪2x d) ಕುಲ೪2x.

42. ☐ ಎಂದಾ ವಿಷಯವು ಸರಿ ☒ ಅಲ್ಲವೇ?

- a) ತ33 b) fiaa c) ್ಝಾಝಾ d) ್ಘಾಘಾ.

43. 9ಪೆಢS ಪೆdd eಢx®'R3ು?

- a) $\square_{\infty, 0\#}$ b) $\vec{s}.fix$ c) ΩY d) $2_{\infty\#}$.

44. 9-' . Rd. 6ð, 3S6 ತS6c6ಂ ರೆಂS□dಂ?
- a) 2ಂಡೆ, b) 2್ಯಸ6ಂ c) ಭೆಂ, S# d) 2್ಯ'À3ಹೆ.
45. ಕೆ3ಕೆಂಸ6 2cಲ 2'ಸ6ಂ-----.
- a) ಭ62aÂ vಂರೆಕ b) ಕೆ3ಕಪ್ಪ c) H2ಲಪ್ಪ,vಂರೆಕ d) 9R6ಪ್ಪ.
46. ಕ&Ñfiog □ಂಡ 2'ಕ'□' -----Hಂಬುd 2ಂಞುj'R
- a) ಕ&Ñ b) ಹು8 c) ಸಪx d) ಹುತ್
47. □ಂಞಂ±ಂಬ6 2fi3 2'ಸ6ಂ
- a) e±ಂಢo b) edŘ c) eಬುÅe d) a±ಂಲ&e.
48. eಂ&fiಂಪ್ಪ,d A&93 ಆd,2 ಂಞ'e'ð 2ಂ&'c, Rಂಞಂ,ð6ಂ
- a) a±ಂಲ&e b) ಕ&ð c) ರಂಕು d) ಹುಚೆಪ್ಪ, 2ಂಞÀ6.
49. □ಂಞಂ 96ತvಂd□'----- ddx3*ಂdಡc 2'ರು'Rಕು.
- a) 8ofi b) 9ofi2 c) fiಂ6ಂ d) &S.
50. 2'cಸುಂಞ3 Are' ಕS3d ಕ9.
- a) u'Roj', b) 89ġ c) ಕು®'ಂಪ□ d) 2ಂಞÀ.

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